

ADVANCED REVIEW

From Traditional Methods to Deep Learning Approaches: Advances in Protein–Protein Docking

 Linlong Jiang¹  | Ke Zhang²  | Kai Zhu¹  | Hui Zhang¹  | Chao Shen³  | Tingjun Hou¹ 

¹College of Pharmaceutical Sciences, Zhejiang University, Zhejiang, Hangzhou, People's Republic of China | ²Polytechnic Institute, Zhejiang University, Zhejiang, Hangzhou, People's Republic of China | ³Department of Clinical Pharmacy, The First Affiliated Hospital, Zhejiang University School of Medicine, Zhejiang, Hangzhou, People's Republic of China

Correspondence: Chao Shen (shenchao513@zju.edu.cn) | Tingjun Hou (tingjunhou@zju.edu.cn)

Received: 17 December 2024 | **Revised:** 17 February 2025 | **Accepted:** 27 February 2025

Editor-in-Chief: Peter R. Schreiner

Funding: This work was supported by the National Natural Science Foundation of China (22377111) and the National Key Research and Development Program of China (2024YFA1307500).

Keywords: deep learning | protein–protein docking | protein–protein interactions

ABSTRACT

Protein–protein interactions play a crucial role in human biological processes, and deciphering their structural information and interaction patterns is essential for drug development. The high costs of experimental structure determination have brought computational protein–protein docking methods into the spotlight. Traditional docking algorithms, which hinge on a sampling–scoring framework, heavily rely on extensive sampling of candidate poses and customized scoring functions based on the geometric and chemical compatibility between proteins. However, these methods face challenges related to sampling efficiency and stability. The advent of deep learning (DL) has ushered in data-driven docking methods that demonstrate significant advantages, particularly boosting the efficiency of protein–protein docking. We systematically review the historical development of protein–protein docking from traditional approaches to DL techniques and provide insights into emerging technologies in this field. Moreover, we summarize the commonly used datasets and evaluation metrics in protein–protein docking. We expect that this review can offer valuable guidance for the development of more efficient protein–protein docking algorithms.

1 | Introduction

Fundamentally, nearly all biological processes within living organisms are executed and regulated through protein–protein interactions (PPIs) or interactions between proteins and other biomolecules [1]. Therefore, determining the three-dimensional (3D) structures of those protein–protein complexes is crucial for elucidating biological pathways and understanding the mechanisms of PPIs. Furthermore, uncovering these complex structures can identify potential pockets and hotspot residues in PPI interfaces, facilitating drug discovery and optimization [2–5]. With the advancement of structural proteomics, various experimental techniques, such as X-ray crystallography, nuclear magnetic resonance

(NMR), and cryo-electron microscopy (cryo-EM), have been used to determine 3D structures of proteins [6, 7]. However, experimentally resolving the structures of protein–protein complexes is more costly and technically challenging compared to individual proteins. Additionally, most structural determination methods capture static snapshots of proteins, lacking dynamic and interaction details [8]. Consequently, relying solely on experimental approaches has significant limitations, prompting researchers to explore computational modeling methods, such as protein–protein docking, as valuable supplements or alternatives [9]. Protein–protein docking exploits information from two or more proteins to determine how protein interfaces could drive the interaction within the most favorable and realistic spatial arrangements.

Traditional protein–protein docking approaches share similarities with those used for protein–small molecule docking [10], primarily utilizing a sampling–scoring framework [11]. This involves a comprehensive conformational search for the ligand protein (the smaller protein) while keeping the receptor protein (the larger protein) stationary, and then scoring functions are applied to estimate and rank the docking scores of the various sampled conformations, facilitating the selection of the final docking pose [12, 13]. However, these approaches often suffer from inefficiency, as they need thousands or even millions of searches to identify potential low-energy conformations [14, 15]. Consequently, there is a growing focus on computational strategies that could effectively balance efficiency and accuracy, notably through the incorporation of deep learning (DL) techniques.

The rapid advancement of DL technologies has been driven by revolutionary progress in hardware, which enables the efficient processing and analysis of increasingly complex data [16, 17]. As a specialized branch of machine learning (ML), DL employs deep neural networks to automatically extract features from large-scale datasets, eliminating the need for the labor-intensive feature engineering typical of traditional ML methods. These neural networks are trained through forward and backward propagation processes, optimizing the relationship between inputs and outputs. In forward propagation, the system computes predictions, while in backward propagation, network parameters are adjusted based on a loss function, refining the ability to predict accurately. By utilizing complex nonlinear activation functions and differentiable structures, DL networks demonstrate remarkable flexibility and capability, making them highly effective at solving a wide variety of complex tasks. With the rapid advancements in computational chemistry and the accumulation of experimental data, the application of DL in the biomedical field has steadily increased, achieving breakthroughs in various downstream tasks, particularly in areas such as medical imaging, drug discovery, and design [18–23].

Currently, computational biologists are leveraging DL for protein complex structure prediction, which can avoid the lengthy search and sampling processes in traditional protein–protein docking and provide direct end-to-end results [24–26]. Specifically, the emergence of convolutional neural networks (CNNs), graph neural networks (GNNs), and diffusion models (DMs) has greatly enhanced the flexibility and diversity of the docking procedure. For example, GNNs have standardized the protein docking task by introducing equivariant and invariant symmetry constraints that align naturally with biological and chemical structural constraints [27]. DMs streamline the complicated sampling phases through efficient pose generation [28], while multimodal inputs for neural networks have enabled the capture of protein conformational changes under different conditions [29–31]. These advances highlight the huge potential of DL in predicting protein structures, establishing it as the preferred approach for modeling proteins and their interactions [32–34].

Up to now, a number of reviews on protein–protein docking methods have been published [11, 13, 35, 36]. However, the majority of these reviews have primarily focused on traditional

approaches, despite the significant attention that DL-based methods have attracted from the research community in recent years. Therefore, in this review, we not only summarize the latest advancements in traditional protein–protein docking strategies but also place a strong emphasis on DL-based docking approaches. We expect that our review will offer valuable insights and guidance for the development of more reliable DL-based protein–protein docking strategies.

2 | Traditional Protein–Protein Docking

Traditional protein–protein docking algorithms consist of two main components: sampling and scoring, which can be employed at different stages of the docking workflow or integrated into a cohesive framework [11]. In a typical docking procedure, the proteins are initially treated as rigid, implying that only their relative locations and orientations are adjusted while their internal structures remain fixed. Specifically, a set of potential poses for the ligand protein is generated based on its physical and geometric compatibility with the receptor protein. Those poses are then scored and ranked using a predefined scoring function, and the top-ranked complex structures may be further refined to account for protein flexibility [37]. Scoring functions can generally be divided into four categories: force-field-based (or physics-based), empirical (or regression-based), knowledge-based (or potential of mean force-based), and descriptor-based (or ML-based) [38, 39]. Given their similarity to the scoring schemes customized for small molecules, we will not discuss these categories in detail here [13]. Due to the large size of proteins, extensive binding interfaces, and often limited binding site information, practical protein–protein docking is often performed as blind docking [36], where docking is conducted without any prior knowledge of the target binding site, making the sampling strategy during the docking process particularly crucial [35]. To enhance sampling efficiency, a variety of search algorithms have been proposed, which can be broadly classified into four categories (Figure 1): (i) exhaustive global search, (ii) local shape feature matching, (iii) randomized search, and (iv) template-based search. Table 1 lists several mainstream protein–protein docking tools and the corresponding web servers. While research has gradually shifted toward DL-based docking strategies, traditional docking search algorithms continue to be actively developed and refined [40–42].

2.1 | Exhaustive Global Search

Exhaustive global search can be employed to explore all potential interaction sites and orientations in protein–protein docking to identify the optimal binding mode. This method typically assumes proteins strictly rigid and uses a grid-based strategy, requiring a comprehensive search across six degrees of freedom (three for translations and three for rotations), implying that the search space grows exponentially, scaling as $O(N^6)$. Enhancing grid resolution or refining Euler angle intervals during the search process can yield higher-resolution docking results but significant increase computational costs. To address this, the fast Fourier transform (FFT) is commonly used as an efficient alternative [43].

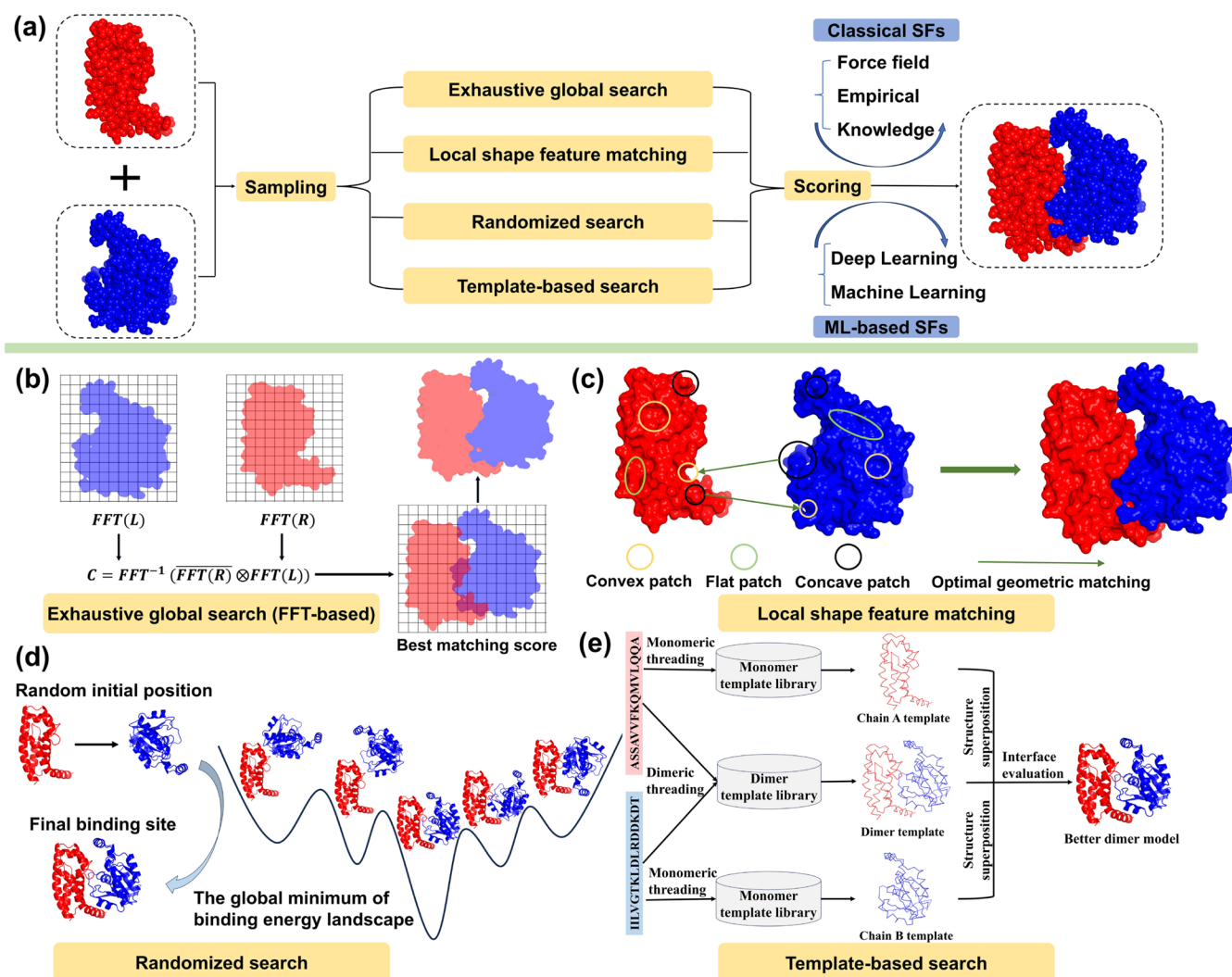


FIGURE 1 | Overview of traditional protein–protein docking. (a) The workflow of traditional protein docking methods, which follows a sampling and scoring framework. (b) An illustration of the most representative exhaustive global search method, the Fast Fourier Transform (FFT) approach, where proteins are mapped onto grids, and the optimal matching score C in the grid is obtained through forward and inverse FFT computations, followed by translational and rotational alignment to generate the final complex. (c) A schematic of local feature matching algorithms, which involve computing molecular surface, detecting the concave, convex, and flat patches using segmentation algorithms, and matching these surface patches to generate the final conformation using geometric algorithms. (d) A diagram of random search algorithms, where the ligand protein is randomly placed in an initial position, and multi-level sampling is performed using random search algorithms to explore the global energy landscape of binding with the receptor protein and identify the global minimum. (e) A schematic of template-based search methods, where the sequence information of the receptor and ligand proteins is obtained, searched separately in a template library to acquire monomer and multimer conformations, and the monomer is subsequently optimized with the multimer to generate the final complex structure.

Once the protein data is discretized into a grid, the FFT search algorithm can be applied, transforming structural information from the spatial domain to a spectral representation in the frequency domain. In this domain, favorable ligand–receptor interactions are rewarded, while overlapping cores are penalized. The results are then converted back to the spatial domain via the inverse Fourier transform. During the conformation search, FFT accelerates the exploration of translational space by iterative computing within discrete intervals. However, since FFT algorithm does not inherently accelerate the rotational space, explicit sampling in this space is required. For each rotation, the discrete match must be recalculated until the entire rotational space is explored, ultimately identifying the optimal docking pose. This FFT-based approach reduces the computational complexity of

both the translational and rotational spaces to $O[N^3 \log(N^3)]$ [43]. The differences among current FFT-based docking programs primarily stem from how protein potentials and information are mapped onto the grid, including factors such as shape complementarity [44, 45], electrostatic energy [46], atomic desolvation effect [47], and hydrophobic complementarity [48], which will lead to different scoring functions. The current FFT searches based on 3D Cartesian grid space are the most popular global search algorithms and have been extensively employed in mainstream docking programs, including FTDock [44], ZDOCK [45, 49], ClusPro [50, 51], HDOCK [52], and so on [53–55].

Exhaustive global search methods can also be enhanced through several acceleration strategies. For example,

TABLE 1 | Mainstream protein–protein docking tools, their features, and web servers.

Docking tool	Search algorithm	Template-based modeling	Residue constraints ^a	Flexible docking	Multimer docking	Web servers
ClusPro [51]	FFT-based search	×	✓	×	✓	https://cluspro.bu.edu/home.php
pyDockWEB [53]	FFT-based search	×	×	×	×	https://life.bsc.es/servlet/pydock/
pyDockSAXS [204]	FFT-based search	×	×	×	×	http://life.bsc.es/pid/pydocksaxs
ZDOCK [205]	FFT-based search	×	×	×	✓	https://zdock.wenglab.org/
HexServer [206]	SFT-based search	×	✓	✓	×	http://hexserver.loria.fr/
FRODOCK2.0 [56]	SFT-based search	×	✓	×	×	http://frodock.chaconlab.org
3D-Garden [207]	Direct search	×	✓	✓	×	http://www.sbg.bio.ic.ac.uk/3dgarden
PatchDock [71]	Local shape matching	×	✓	×	×	http://bioinfo3d.cs.tau.ac.il/PatchDock/
SymmDock [71]	Local shape matching	×	×	×	✓	http://bioinfo3d.cs.tau.ac.il/SymmDock/
RosettaDock [198]	Monte Carlo-based search	×	✓	×	×	https://rosie.graylab.jhu.edu/
HawkDock [208]	Monte Carlo-based search	×	✓	×	×	http://cadd.zju.edu.cn/hawkdock/
SwarmDock [209]	Particle Swarm Optimization	×	✓	✓	×	http://bmm.cancerresearchuk.org/%7ESwarmDock/
HADDOCK [91]	Simulated annealing	×	✓	✓	×	http://haddock.chem.uu.nl/services/HADDOCK
FoXSDock [210, 211]	SAXS-based modeling ^b	×	✓	✓	×	http://modbase.compbio.ucsf.edu/foxsdock/
InterEvDock2 [212]	Template-based and SFT-based search	✓	×	×	×	https://bioserv.rpbs.univ-paris-diderot.fr/services/InterEvDock2/
GRAMM-X [213, 214]	Template-based and FFT-based search	✓	✓	×	×	https://gramm.compbio.ku.edu/
HDOCK [52]	Template-based and FFT-based search	✓	✓	×	✓	http://hdock.phys.hust.edu.cn/

^aThis column specifies whether the protein docking tool allows certain residues to be designated as binding sites during docking.^bThis method refers to modeling proteins using small-angle X-ray scattering (SAXS) data of protein complexes.

FRODOCK [56, 57] and Hex [58, 59] use a rotational search based on spherical harmonics or spherical polar Fourier correlations (SFT) in the 3D Cartesian grid space to improve search efficiency. SOFTDOCK [60] and SKEDOCK [61, 62] employ Boolean operators and heuristic rules to directly identify

regions with matching shapes in the 3D grid, facilitating geometric fitting between two proteins. Additionally, some approaches benefit from advanced software and hardware techniques, such as GPU acceleration and convolution libraries in Cell-Dock [49, 63, 64].

2.2 | Local Shape Feature Matching

Local shape feature matching sampling identifies and aligns highly complementary binding interfaces between two proteins by recognizing and matching the surface's local geometric features. This approach utilizes mathematical tools or algorithms, such as surface geometric descriptors or shape distributions [65], to capture and analyze local surface features, such as accessible surface areas, curvature, protrusions, and depressions [66]. By comparing these descriptors, it identifies complementary shapes, ensuring precise alignment of matching features. Once potential matching features are identified, the method rapidly generates various binding orientations, each representing a possible protein conformation based on relative position and orientation. These conformations are then evaluated based on shape complementarity, electrostatic compatibility, and/or potential energy calculations, with the highest-ranked ones considered most likely to reflect true biological interactions [65].

By matching local features rather than entire protein surfaces, docking simulations avoid the exploration of all six degrees of freedom in global space. Instead, 3D translational and 3D rotational parameters are implicitly embedded in the matching transformation matrix, which is calculated only when a match is found, thereby significantly reducing computational load [67, 68]. Nevertheless, the binding orientations produced by these algorithms often contain atomic clashes or redundant conformations. To address this, stereochemical checks are routinely conducted to eliminate solutions with excessive clashes, followed by a clustering step to remove redundancies from the final set of solutions [69].

One prominent docking program that utilizes local shape feature matching is DOCK [70], which employs a distance geometry algorithm for matching and subsequently performs local conformational searches anchored to these matches, often completing the docking process within minutes. Another example is PatchDock [71], which uses geometric hashing and pose clustering techniques to match interactions between two proteins based on geometric features of surface fragments, such as concave, convex, and flat segments. Similar docking algorithms in this category include SnapDock [72] and LZerD [73, 74]. GAPDOCK [75], however, represents proteins using point surfaces calculated by the Connolly program [76] and applies a genetic algorithm to optimize surface complementarity between two proteins by moving one protein's surface closer to the other.

2.3 | Randomized Search

Randomized search methods offer an effective strategy for exploring broad search spaces. Unlike exhaustive global search, these methods do not require specific protein representations, such as grids or shapes. Instead, they facilitate direct atomic-level searches or the incorporation of coarse-grained models. In random search, ligand proteins are randomly placed around the binding site for local searches or distributed across the entire receptor for global searches, each with a randomly assigned initial pose [77, 78]. Initially, protein shape or surface characteristics guide the optimal placement of the ligand protein, generating

more plausible initial binding orientations. Subsequently, a multi-stage sampling approach utilizing random algorithms refines the relative orientation of the two proteins. Compared to the exhaustive search's requirement of adjusting six degrees of freedom, random algorithms tune the corresponding parameters from the starting pose, markedly decreasing computational costs. Furthermore, multi-stage conformational sampling can achieve global energy optimization with fully flexible side chains to induce conformational changes in ligand-binding residues [79–81].

Common random search methods include Monte Carlo algorithms [81–84], genetic algorithms [75, 85], and simulated annealing [86, 87]. Docking algorithms based on these methods are outlined below. RosettaDock [82] typically executes local docking by randomly placing ligands at the binding site and optimizing the binding conformations through a multi-stage Monte Carlo process. For flexible docking, it employs a low-resolution search followed by side-chain repacking or the sampling of external side-chain conformations of rotamers, yielding flexible docking conformations. In contrast, ATTRACT [83] performs global docking using pseudo-atoms to represent protein residues and generate multiple starting positions on the surface of the receptor protein, all of which undergo energy minimization via a Monte Carlo search. For flexible docking, ATTRACT also accommodates side-chain flexibility by combining a rotamer library with replica exchange methods. SwarmDock [88] and LightDock [89] adopt the Particle Swarm Optimization (PSO) algorithm for sampling, where ligands are represented as a swarm orbiting the receptor. Each particle's position and velocity are updated based on the swarm's best position, enabling the exploration of the flexibility of interacting partners after binding. FPDock [90] utilizes the Flower Pollination Algorithm (FPA), starting with a randomly generated initial population that seeks to find local optima. Each “pollination” cycle attempts to improve scores by combining with the best or randomly chosen “pollen”. The binding results are evaluated using an energy-based scoring function that integrates the contributions of electrostatic potential and van der Waals forces. HADDOCK [86, 91, 92] guides the docking process with ambiguous interaction restraints and leverages simulated annealing for energy minimization, ensuring optimized binding conformations while accounting for both interface backbone and side-chain flexibility.

2.4 | Template-Based Search

In the protein–protein docking field, another important strategy is template-based docking methods [93–95]. These methods rely on the structures of known protein complexes to predict the interactions between new protein pairs, operating under the assumption that proteins with similar sequences or structural motifs will share comparable interaction patterns. Thus, the accuracy of the predicted structures depends on the evolutionary relationship between the target and template proteins in the template library, as well as the accessibility of reliable templates therein [41]. The strategies of different methods primarily diverge in their techniques for detecting evolutionary information between target and template proteins, which can be classified into three categories: (i) sequence–sequence alignment [96–98], (ii) sequence–threading

[99–101], and (iii) structural alignment [102–106]. Sequence-based alignment typically requires a high degree of sequence similarity between the target and its corresponding template. Conversely, when the sequence similarity among homologous samples is low, structural alignment generally delivers better accuracy.

Template-based docking methods predominantly utilize experimentally determined protein complex structures as templates, primarily sourced from the Protein Data Bank (PDB) [107] database. To identify suitable templates, sequence alignment tools like BLAST [108, 109] or structure alignment tools like TM-align [110] are typically used to compare target protein sequences and structures with known complexes in the database [100, 111]. Upon identifying potential templates, their structures are used to infer possible binding modes for the target proteins. This step determines the quality and accuracy of the docking prediction. The selected templates guide the initial docking process, where the target proteins are aligned according to the template structures, thereby forming a starting conformation that closely approximates the native complex. To improve this initial docking and enhance accuracy, computational techniques such as energy minimization and molecular dynamics (MD) simulations are employed to refine the interface and eliminate any spatial clashes or unfavorable interactions [112].

Given their reliance on experimentally determined structures, these methods provide highly confident predictions when the template closely matches the target proteins. Using existing structures as a foundation, template-based methods can significantly reduce the computational resources and time required compared to the sampling and scoring strategies previously discussed. Nonetheless, their applicability is constrained by the availability of homologous protein complexes and the conformational changes at interaction sites due to sequence differences. As the number of resolved protein complexes increases and multi-template methods are improved, template-based docking methods will gain increasing value [113–115].

3 | Deep Learning Methods

Traditional docking algorithms based on the sampling and scoring strategy still require extensive search and evaluation processes despite employing various optimization techniques. These results in high computational complexity, and require considerable resources and time, and together these features make it difficult to achieve a global optimal solution. Furthermore, these traditional methods rely on empirical scoring functions to evaluate the stability and energy of docking poses. However, these scoring functions exhibit limitations in both accuracy and generalizability, making it challenging to accurately model complex biomolecular interactions. To overcome these challenges, DL methods have been introduced into this field. By leveraging the powerful representation learning capabilities of neural networks, DL approaches can automatically extract features and patterns from extensive protein datasets, thereby enhancing both the accuracy and efficiency of docking [116–118]. Table 2 outlines current DL methods

applied to protein–protein docking, primarily focusing on two areas: rigid protein–protein docking based on structural information and protein complex structure prediction based on sequence data and protein co-folding.

3.1 | Rigid Protein Docking

Solving the rigid protein–protein docking problem involves learning the rigid-body transformation from two unbound structures to their bound state while keeping their conformations unchanged throughout the process. This assumption simplifies protein–protein docking to six degrees of freedom compared to flexible docking, which requires considering the degrees of freedom of each atom, substantially increasing computational demands. Often, protein conformational changes during docking are not significant, making the rigid-body assumption a reasonable approximation for many practical applications [119–122]. Thus, DL for rigid protein docking tasks can be formulated as:

$$R, \vec{t} = \phi \left[\vec{G}_1(\vec{X}_1, H_1); \vec{G}_2(\vec{X}_2, H_2) \right] \quad (1)$$

where $\vec{G}$ represents the protein, $\vec{X}$ is the coordinates of the protein, H is the features of the protein, ϕ is a prediction function, R is the output rotation matrix, and $\vec{t}$ is the output translation vector. As shown in Figure 2, current DL strategies for rigid protein docking can be divided into the following three types: (i) protein surface feature learning and complementarity, (ii) coordinate updates based on equivariant graph neural network (EGNN), and (iii) sampling and generation based on DM.

3.1.1 | Protein Surface Feature Learning and Complementarity

Early attempts in applying DL to protein docking primarily focus on utilizing geometric deep learning (GDL) to model and represent protein surfaces based on their structural and chemical characteristics [123, 124]. These approaches initially use neural networks to predict inter-protein contacts based on binding site interfaces, incorporating both structural features and potential evolutionary characteristics of the input proteins. The resulting predicted binding sites or inter-chain contact maps can then guide the docking process, ultimately leading to the determination of the final complex structure [125–127]. This DL-driven strategy reframes the protein docking task as a puzzle-solving problem, mirroring the traditional local shape feature matching approach and segmenting it into three steps: (i) puzzle generation, where DL techniques are applied to model the protein surface, (ii) puzzle matching, where surface matching occurs between two proteins, and (iii) puzzle evaluation, where potential docking poses are evaluated and ranked using either DL algorithms or physics-based energy functions.

The initial puzzle generation phase is particularly crucial, highlighting the importance of surface learning and protein modeling. Protein surface representation is an advanced description of protein shape, significantly minimizing

TABLE 2 | Summary of DL-based protein–protein docking methods.

Model	Year	Model type	Input ^a	Use MSAs ^b	Applications	Code availability
dMaSIF [25]	2022	Surface learning	Structure	No	Rigid protein docking/ protein interface prediction	https://github.com/FreyrS/dMaSIF
HMR [129]	2023	Surface learning	Structure	No	Rigid protein docking/ binder pocket classification	https://github.com/bytedance/HMR
EquiDock [24]	2021	EGNN	Structure	No	Rigid protein docking	https://github.com/octavian-ganea/equidock_public
HSRN [140]	2022	EGNN	Structure/sequence	No	Rigid protein docking/ antibody design	https://github.com/wengong-jin/abdockgen
SyNDDock [143]	2023	EGNN	Structure	No	Multimeric protein docking	/
ElliDock [141]	2024	EGNN	Structure	No	Rigid protein docking	https://github.com/yaledeus/ElliDock
EBMDock [142]	2024	EGNN/energy-based	Structure	No	Rigid protein docking	https://github.com/wuhuaijin/EBMDock
DiffDock-PP [26]	2023	DM	Structure	No	Rigid protein docking	https://github.com/ketatam/DiffDock-PP
DiffMasif [148]	2023	DM	Structure	No	Rigid protein docking	/
LatentDock [149]	2023	DM	Structure	No	Flexible protein docking	/
DOCKGAME [150]	2023	DM/energy-based	Structure	No	Multimeric protein docking	https://github.com/vsomnath/dockgame/
DFMDock [151]	2024	DM/energy-based	Structure	No	Rigid protein docking	https://github.com/Graylab/DFMDock
AlphaFold-Multimer [33]	2022	Sequence-based	Sequence	Yes	Protein folding	https://github.com/deepmind/alphafold
ColabFold [162]	2022	Sequence-based	Sequence	Yes	Protein folding	https://github.com/sokrypton/ColabFold
Uni-Fold MuSSe [166]	2023	Sequence-based	Sequence	No	Protein folding	/
DMFold-Multimer [165]	2024	Sequence-based	Sequence	Yes	Protein folding	https://zhanggroup.org/DeepMSA/
AlphaFold3 [34]	2024	Sequence-based	Sequence	Yes	Protein folding	https://github.com/google-deepmind/alphafold3
MolPC [171]	2022	Multi-step	Sequence	Yes ^c	Multimeric protein docking	https://colab.research.google.com/github/patri-ckbryant1/MoLPC/blob/master/MoLPC.ipynb
Alpha-RED [170]	2023	Multi-step	Sequence	Yes ^c	Flexible protein docking	https://github.com/Graylab/AlphaRED
CombFold [172]	2024	Multi-step	Sequence	Yes ^c	Multimeric protein docking	https://github.com/dina-lab3D/CombFold
xTrimDock [173]	2023	Multimodal fusion	Sequence/structure	No	Rigid protein docking	/

(Continues)

TABLE 2 | (Continued)

Model	Year	Model type	Input ^a	Use MSAs ^b	Applications	Code availability
GeoDock [31]	2023	Multimodal fusion	Sequence/structure	No	Flexible protein docking	https://github.com/Graylab/GeoDock
DockGPT [30]	2023	Multimodal fusion	Sequence/structure	No	Flexible protein docking	https://github.com/MattMcPartlon/protein-docking

^aOnly sequence and structure inputs are considered; 'Sequence/Structure' indicates that both are required according to different downstream applications.

^bIndicate whether the model utilizes MSAs.

^cThese multi-step integration methods all utilize MSAs when performing structure prediction using the AF2 architecture.

redundancy when compared to full-atom or residue-level representations of 3D structures. Surface learning not only deciphers the interactions between interfaces but also facilitates the subsequent matching process. Techniques for protein surface learning primarily include grid-based neural networks [123], point-cloud-based neural networks [128], and manifold-based neural networks [129, 130]. The matching stage generally depends on the features of the two protein surfaces, employing mechanisms such as cross-attention to predict binding interfaces, followed by rigid alignment for structural transformation.

In 2020, Gainza et al. introduced MaSIF, pioneering the use of GDL to represent protein surfaces for downstream applications [123]. However, MaSIF requires extensive manual feature extraction and heavily relies on precomputed grids, resulting in substantial computational and memory demands. To overcome the limitations of grid-based networks, Sverrisson et al. proposed dMaSIF, marking an early attempt at using point-cloud neural networks [128]. In dMaSIF, the protein surface is sampled to gather representative points, and feature extraction is applied to capture spatial position, normal vector, curvature, and chemical characteristics. The network then employs convolution mechanisms to learn the relationships between neighboring points on the protein surface, effectively extracting surface features.

$$h'_i = \sum_{j \in N(i)} (x_i - x_j) \cdot \text{MLP}(h_i, h_j, \|x_i - x_j\|) \quad (2)$$

where i represents a sampled point on the protein surface, j denotes a neighboring node adjacent to i , x and h refer to the coordinates and feature vectors of the nodes, MLP updates the features, and h' represents the final learned surface point features. This point-cloud-based approach does not depend on predefined topological structures, enabling detailed representation of protein surface geometry and making it well-suited for handling complex and irregular protein surfaces. Point clouds inherently support SE (3) equivariance, avoiding the need for complex coordinate transformations and offering greater flexibility compared to traditional grid or voxel representations.

In 2023, Wang et al. proposed HMR [129], which utilizes Riemannian manifold-based GDL to capture protein surface shapes, representing the chemical properties of protein surfaces through functions distributed across the manifold. Under this representation, each protein is assigned a unique "shape-DNA" defined as a set of eigenvalues from the Laplace–Beltrami operator on the protein surface:

$$\Delta_M \vartheta_i = \lambda_i \vartheta_i \quad (3)$$

where Δ is the Laplace–Beltrami operator, ϑ_i is the i th eigenfunction, and λ_i is the corresponding eigenvalue. Harmonic analysis is then performed using the eigenfunctions ϑ_i , expanding a function on the manifold as a linear combination of a series of orthogonal basis functions, allowing for multi-resolution feature representation at each point on the protein surface. This enables concise representation of different chemical properties as a set of linear combination coefficients. Given a point on the protein surface, its feature representation and message passing are as follows:

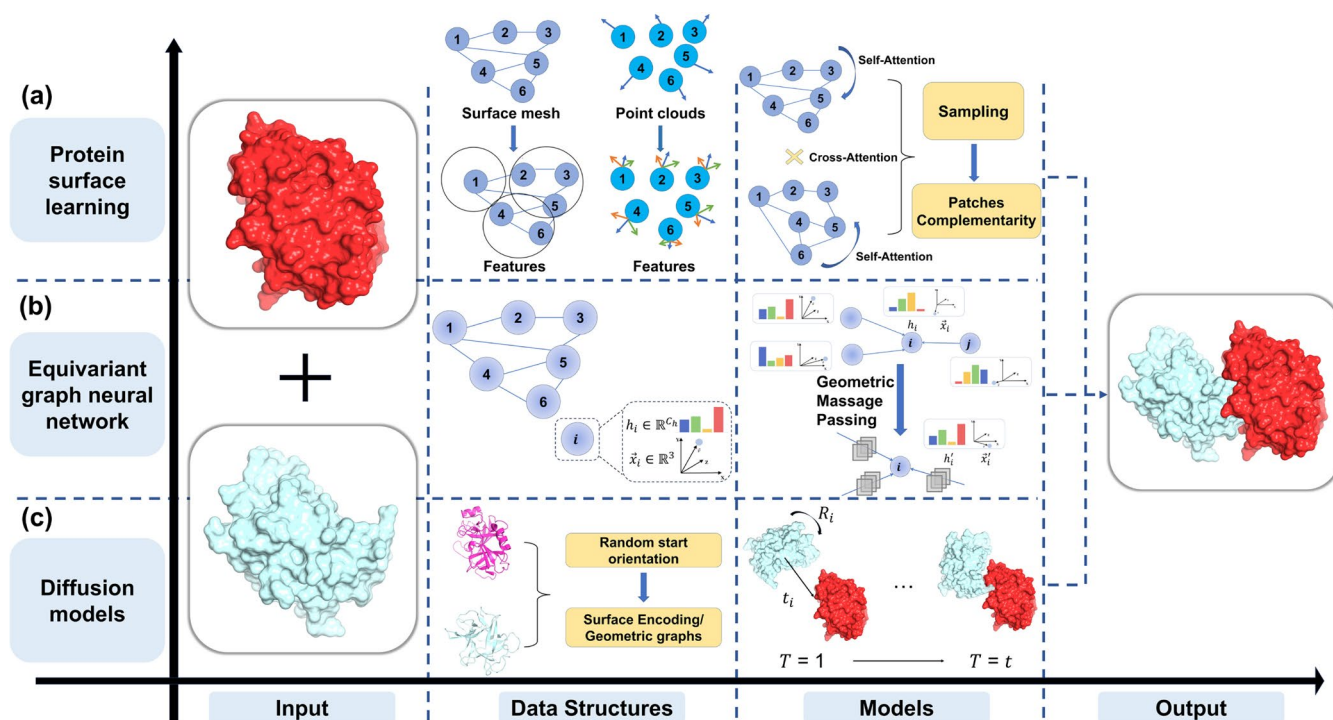


FIGURE 2 | Illustration of the complete input-output pipeline for deep learning-based rigid protein docking, spanning data structures and models. (a) Protein surface learning and matching: Input protein structures are processed to learn surface features using methods like point cloud or manifold neural networks. Surface features of receptor and ligand proteins are matched via cross-attention mechanisms to output the final complex structure. (b) Coordinate updates based on EGNN: Protein structures are represented as graphs, and EGNN modules are used to update coordinates, generating translation and rotation matrices for assembling the complex. (c) Sampling and generation based on DM: The ligand protein is randomly initialized, and iterative forward noise addition and reverse denoising processes predict ligand positions. A discriminative model selects the final complex conformation.

$$h(x) = [\theta_1(x), \theta_2(x), \dots, \theta_k(x)] \quad (4)$$

$$h_v^{k+1} = \text{AGGREGATE}\left(\left\{h_v^k \cdot \theta_i(e_{uv}) \mid u \in N(v)\right\}\right) \quad (5)$$

where h_v^k is the feature of node v at the k th layer, $N(v)$ is the set of neighbors of node v , e_{uv} is the edge feature between nodes u and v , and AGGREGATE denotes a function that aggregates neighbor features. A readout function then aggregates the features of all nodes. The Laplace–Beltrami operator generates eigenvalues and eigenfunctions that capture the curvature, shape, and other geometric features of the molecular surface. Additionally, harmonic eigenfunctions are inherently invariant to rotation and translation, avoiding biases introduced by complex coordinate transformations.

After generating puzzle pieces, the next essential step is to align and evaluate them to facilitate protein–protein information exchange, leading to a variety of innovative strategies. Soltanikazemi et al. used inter-residue contacts as distance constraints to facilitate information transfer between proteins [125]. Sverrisson et al. employed a cross-attention mechanism to calculate binding energy matrices for all possible surface point pairs, transforming the energy matrix into a probability distribution over point pairs [25]. This distribution was then used to generate candidate docking conformations, which were subsequently ranked and evaluated by a discriminative model informed by

prior physical knowledge to assess binding positions. Wang et al., on the other hand, framed puzzle matching as a binary classification problem, employing cross-attention to classify each point-pair feature based on its probability of belonging to the binding interface. Subsequently, they used functional maps to align the corresponding point pairs between receptor and ligand surfaces, ultimately applying rotations and translations to the ligand protein to achieve the final complex structure [129].

3.1.2 | Coordinate Updates Based on EGNN

This type of approach treats protein–protein docking as a regression problem by directly using EGNN to predict the movement of the ligand protein relative to the receptor protein and updating the corresponding coordinates through message passing. The core concept here is to exploit equivariance to ensure that the predicted results remain consistent under identical rotation and translation operations across different input coordinate frames. This requires capturing equivariance through EGNN, as traditional GNNs cannot achieve equivariance across varying coordinate transformations [131, 132]. EGNN has been applied to various 3D modeling tasks, including atomic potential prediction [133, 134], molecular property prediction [135, 136], and protein–ligand docking [137, 138]. Specifically, in the realm of protein–protein docking, EGNN has enabled fast, end-to-end prediction of docking poses between ligand and receptor proteins.

Unlike invariance, equivariance requires that when the input data undergoes a transformation, the network's output must transform accordingly, thereby preserving the consistency of its intrinsic structure and features. In rigid protein–protein docking, equivariance is particularly emphasized during geometric transformations between ligand and receptor proteins [24, 27]. Given ligand protein as X_1 and receptor protein as X_2 , with Q and g denoting their respective rotations and translations, the objective of rigid-body protein–protein docking is to dock $Q_1X_1 + g_1$ onto $Q_2X_2 + g_2$. The rotation matrix $R(Q_1X_1 + g_1 | Q_2X_2 + g_2)$ and translation vector $t(Q_1X_1 + g_1 | Q_2X_2 + g_2)$ are obtained through a systematic transformation process from the original $R(X_1 | X_2)$ and $t(X_1 | X_2)$. The specific transformation process includes (i) applying Q_1^T inversely to reverse the rotation Q_1 of X_1 , thereby restoring the original transformation of the ligand; (ii) applying $R(X_1 | X_2)$ to the ligand to execute the original docking rotation; and (iii) applying Q_2 to the docked ligand and receptor to obtain the final rotation of the docked complex. The resulting rotation matrix and translation vector are as follows:

$$R(Q_1X_1 + g_1 | Q_2X_2 + g_2) = Q_2R(X_1 | X_2)Q_1^T \quad (6)$$

$$t(Q_1X_1 + g_1 | Q_2X_2 + g_2) = Q_2t(X_1 | X_2) - Q_2R(X_1 | X_2)Q_1^Tg_1 + g_2 \quad (7)$$

For any rotation $Q_1, Q_2 \in SO(3)$ and translation $g_1, g_2 \in \mathbb{R}^3$ satisfying the above equations, the SE(3) equivariance of the predicted docking complex is guaranteed. Models that satisfy these equations ensure that any SE(3) transformation applied to X_1 is equivariant, and similarly for transformations applied to X_2 .

Equidock stands out as a highly influential DL model for rigid protein–protein docking, pioneering the integration of the EGNN framework. It utilizes an SE(3) graph matching network and an optimized transport matching algorithm to ultimately compute the rotation and translation of rigid ligand proteins [24]. Initially, proteins are represented as k-nearest-neighbor (k-NN) graphs based on Euclidean distances between the nodes, with nodes denoting amino acids and features encompassing amino acid types and geometric positions. Then, EGNN is used to learn the feature representations. For each pair of adjacent nodes, i and j , it computes the messages m_{ij} and μ_{ij} transmitted from node j to node i . Finally, the information from neighboring nodes is aggregated to update features, ensuring the co-transformation of node coordinates and features while maintaining equivariance. The specific calculation formulas are as follows:

$$m_{ij} = \varphi^e(h_i^{(l)}, h_j^{(l)}, \|x_i^{(l)} - x_j^{(l)}\|^2, e_{ij}) \quad (8)$$

$$a_{ij} = \frac{\exp\{\left[\psi^q(h_i^{(l)}), \psi^k(h_j^{(l)})\right]\}}{\sum_{j'} \exp\{\left[\psi^q(h_i^{(l)}), \psi^k(h_{j'}^{(l)})\right]\}} \quad (9)$$

$$\mu_{ij} = a_{ij}Wh_j^{(l)} \quad (10)$$

$$x_i^{(l+1)} = x_i^{(l)} + \sum_{j \neq i} (x_i^{(l)} - x_j^{(l)})\varphi^x(m_{ij}) \quad (11)$$

$$h_i^{(l+1)} = \varphi^h\left(h_i^{(l)}, \sum_{j \in N(i)} m_{ij}, \sum_{j \in N(i)} \mu_{ij}\right) \quad (12)$$

where h represents node features, x represents node coordinates, e represents edge features, $N(i)$ denotes the set of neighbors of node i , a is an attention-based coefficient for weighting messages from different neighbors, W is a learnable matrix, and $\varphi^e, \varphi^x, \varphi^h, \psi^q, \psi^k$ are output vectors with the parameters of φ shared or varied across layers.

Given the multiple potential binding sites on protein surfaces, predicting the docking pose accurately requires identifying key points on each protein and minimizing matching costs through optimal transport. A multi-head attention mechanism is employed to extract key points from ligand and receptor graphs, organizing their coordinates into two point sets. A matching matrix is then defined to describe the correspondence between these two sets, with the matrix satisfying two constraints that ensure one-to-one matching and non-negative elements. The goal is to find the matching matrix that minimizes transport costs. Finally, a differentiable variant of the Kabsch algorithm [139] is adopted to align the two proteins, outputting the rotation matrix R and translation vector $\vec{t}$ to predict the final blind docking pose.

There are also methods that achieve equivariance to spatial rotations and translations without directly relying on the EGNN framework by utilizing similar coordinate update mechanisms. Jin et al. introduced the Hierarchical Structural Refinement Network (HSRN) [140], which applies the principle of protein equivariance to antigen–antibody docking and optimization. HSRN adopts a hierarchical pseudo-force prediction method that predicts relative interatomic forces rather than directly calculating absolute coordinates. The network iteratively refines docking interfaces using residue-level and atom-level graphs: the residue graph facilitates global structural adjustments, while the atom graph refines side-chain interactions. Through continuously updating the antibody structure, HSRN effectively adapts it to the antigen epitope.

Additionally, integrating the EGNN strategy with protein surface learning can yield enhanced docking results. For example, Yu et al. introduced ElliDock [141], which represents the protein–protein docking interface using an elliptic–paraboloid surface. The elliptic–paraboloid form was chosen for clearly defined, closed-form SE(3) transformation properties in 3D space, thus avoiding complex redundant protein information. The approach also employs an EGNN-derived framework known as EPIT, which uses an equivariant message-passing network as its backbone, augmented by graph transformer layers to learn features adapted to protein ellipsoid surfaces. By calculating the overlap transformation of two ellipsoid–paraboloids under equivariance, the approach can efficiently and accurately accomplish protein–protein docking.

Other approaches, represented by EBMDOck [142] developed by Wu et al. and SyNDock [143] proposed by Ji et al., rely solely on EGNN for protein feature learning but employ alternative strategies for subsequent processing. EBMDOck starts with an EGNN framework to extract protein features and further introduces an SE(3) equivariant cross-attention module to aggregate ligand–receptor interactions. These features are then combined with a mixture density network to formulate an energy function. Then, Langevin dynamics sampling is applied to optimize initial docking sites, selecting the site with

the lowest energy from multiple samples as the final predicted docking pose. On the other hand, SyNDock aims to address the challenge of multi-body rigid protein–protein docking by reformulating it as a problem of learning global transformations. It also uses the EGNN architecture to extract features from protein graphs, estimating the relative transformations between protein chain pairs and their corresponding confidence values. However, it further incorporates a differentiable transformation synchronization module to recover global absolute transformations, thereby ensuring the accuracy of the predicted multi-body complex.

Nevertheless, these approaches, which frame protein–protein docking as a regression problem, aimed at directly predicting the final pose through a rotation matrix and translation vector, are limited to generating a single prediction result [144]. This limitation distinguishes them from traditional methods, which generate multiple docking conformations primarily as a strategy to increase the likelihood of capturing the correct binding mode. Consequently, such models are more susceptible to errors in blind docking scenarios, resulting in reduced accuracy in final predictions.

3.1.3 | Sampling and Generation Based on DM

To avoid the issue of deterministic models that yield a single binding conformation, DM describes protein–protein docking as a generative problem. This approach captures a broad spectrum of candidate conformations, leading to more reasonable, accurate, and diverse complex structures. Given two unbound proteins, DM has the capability to estimate the distribution of all potential ligand–protein poses, and then a discriminator-based confidence model evaluates these poses, ultimately selecting the highest-ranked docking pose as the final prediction. This strategy resembles traditional docking algorithms that follow a sampling–scoring architecture but replaces traditional sampling with a DM paradigm.

DM consists of a forward diffusion process and a reverse generative process [145–147]. Taking the denoising diffusion probabilistic model (DDPM) as an example, it serves as a practical instantiation of the broader DM framework. While DM gradually adds noise in the forward process and aims to reconstruct the original distribution in the reverse process, DDPM formalizes this by assuming that both processes follow a Markov chain structure. In DDPM, noise is added in a controlled manner during the forward process, and the reverse process gradually removes this noise. During the forward diffusion process, starting from an initial conformational distribution $x_0 q(x)$, Gaussian noise is gradually introduced. The variance of this noise is determined by a fixed value β_T within the interval $(0, 1)$, and its mean is determined by both β_T and the current state x_t at time t . As time step t increases ($t \rightarrow T$), the conformational distribution gradually transforms into an independent Gaussian distribution. The process can be described as:

$$x_t \sim q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t} x_0, (1 - \alpha_t)I) \quad (13)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

On the other hand, the reverse generative process $p(x_{t-1} | x_t)$ involves denoising. At each time step t , a random two-dimensional Gaussian noise is initially sampled and then progressively denoised, generating a structure x_0 that conforms to the distribution of real conformations. However, since the reverse diffusion process $p(x_{t-1} | x_t)$ is unknown, we approximate it with a parameterized model $p_\theta(x_{t-1} | x_t)$. DDPM assumes that the reverse denoising process also removes Gaussian noise. In theory, a random Gaussian noise is determined by parameters mean μ_θ and variance Σ_θ , so $p_\theta(x_{t-1} | x_t)$ can be described as follows:

$$p_\theta(x_{t-1} | x_t) = N\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right) \quad (14)$$

Simultaneously, in reasoning through the forward diffusion process of DDPM, it is assumed that each time diffusion follows a Markov process. The backward process of the DM can be described by the following Markov chain:

$$p_\theta(x_{0:T}) = p(x_T) \cdot \prod_{t=1}^T p_\theta(x_{t-1} | x_t) \quad (15)$$

where $p(x_T) = N(x_T; 0, I)$.

Building upon the sophisticated framework of DiffDock [144] for small-molecule docking, Ketata et al. proposed DiffDock-PP [26], which stands as the earliest application of DM to rigid protein–protein docking. The model initially represents two unbound proteins as a heterogeneous geometric graph. Through a forward diffusion process, noise is incrementally added to the input graph, gradually transforming the ligand protein structure into increasingly noisy intermediate states. Starting from these noisy intermediates, the reverse denoising process is applied, using score matching to refine the translations and rotations of the ligand protein, thereby generating multiple potential docking conformations. An SE(3) equivariant convolutional network is then employed as a confidence model to evaluate the quality of structures sampled from the DM. By iterative sampling, calculating L-RMSD values, and comparing them to a predefined threshold, a cross-entropy loss is derived to guide the training of the confidence model. During the inference stage, a set of candidate poses can be sampled from the DM and ranked according to the trained confidence model, ultimately producing high-confidence poses.

The strategy of DM can also be integrated with surface learning. DiffMaSIF [148] leverages the strengths of both protein surface learning and the score-based diffusion process. This model incorporates a geometric pretraining approach, analyzing protein surface point clouds to learn point-set, local, and global geometric features, which are then transformed into feature vectors by the encoder. During the diffusion process, protein molecular motion is modeled on a product manifold of 3D translations and rotations. Vector neurons and SE(3) equivariant graph convolutions are then utilized to produce translation and rotation scores, guiding the rotational and translational alignment of docking proteins. This culminates in scored configurations that enable high-precision docking predictions.

LATENTDock [149] pioneers the use of a latent diffusion model by incorporating a latent diffusion process within the latent space of a pretrained autoencoder. Initially, the autoencoder is trained to map proteins to a rotation- and translation-invariant latent space. Noise is progressively added to the protein representations until they conform to a uniform noise distribution. In the reverse diffusion process, noise is iteratively removed, allowing the decoder to reconstruct the protein complex structure from its latent representation. LATENTDock also supports conditional docking based on specific binding site information and accommodates backbone and side-chain conformational changes, offering a level of docking flexibility.

Certain strategies use neural networks to approximate the potential energy function of training data and utilize diffusion models to sample docking conformations. One of the most innovative aspects of DOCKGAME [150] is the training of energy functions either supervised by physics-based models or self-supervised through denoising score matching. During the diffusion process, DOCKGAME samples from the Gibbs distribution of the true potential by learning the DM across the action spaces (rotations and translations) of proteins. Additionally, this approach emphasizes multi-body protein docking, marking the first application of DM in this context. DFMDock [151] is well known for its seamless integration of a diffusion sampling model with a confidence ranking mechanism. This approach is designed with dual output heads: one for force prediction and the other for energy estimation. The model is trained by aligning denoised forces with target

values while energy gradients are optimized to correspond with these forces. This design enables the model to generate samples guided by predicted forces and rank conformations based on predicted energy scores, thereby obviating the need for a separate confidence model.

However, while diffusion models facilitate the generation of a wider range of docking regions and poses, they come with a trade-off at the expense of computational efficiency compared to deterministic models.

3.2 | Flexible Protein Docking

Rigid protein docking operates under the assumption that proteins remain rigid during the binding process, neglecting solvents and ion effects, and completely ignoring the flexibility of proteins in their bound forms, which severely limits their applicability. The ultimate goal of protein docking is not only to ensure the alignment of binding interfaces but also to account for the flexibility and conformational changes in both the protein backbone and side chains. Consequently, flexible protein docking becomes indispensable, as it accurately captures the dynamic conformational adjustments during binding, offering more biologically meaningful insights [29, 152]. As shown in Figure 3, current strategies for flexible docking can be summarized into three types, including sequence-based structure prediction approaches, multi-step integration approaches, and docking approaches that rely on multimodal fusion architectures.

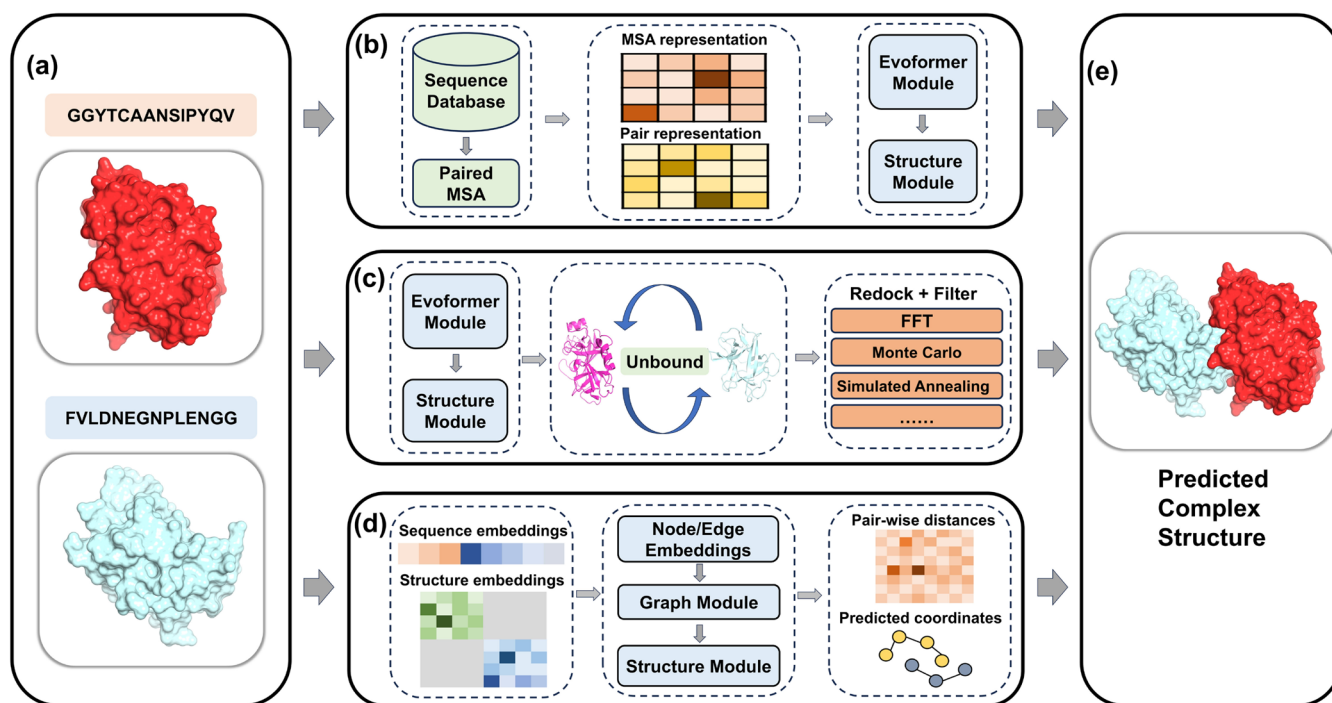


FIGURE 3 | Illustration of DL-based flexible protein docking. (a) The model input can be structural data, sequence data, or a combination of both. (b) Direct structure generation from sequence: The input sequence undergoes MSAs to extract pairwise information, which is then fed into the Evoformer module and structure module to generate conformations. (c) Multi-step approach for conformation generation: Initially, protein complex structures are generated based on protein folding. The receptor and ligand are then separated for re-docking and filtering using traditional algorithms. (d) Multimodal fusion architecture for prediction: Features from protein structure and sequence are integrated using PLMs and GNNs to predict residue positions. (e) The final complex structures often exhibit significant flexibility and conformational changes.

3.2.1 | Sequence-Based Structure Prediction

Shifting focus from protein structure to sequence offers a pathway to overcome the challenges associated with the inherent rigidity and dynamic nature of proteins. Traditional structure-based methods often struggle with these inherent characteristics. In contrast, sequence-based approaches leverage the information encoded within protein sequences to iteratively assess the physical plausibility of predicted structures. Through a rigorous optimization process, these methods refine predicted coordinates, gradually converging toward the final complex conformation. This approach not only circumvents the limitations of relying solely on static structures but also illuminates the flexibility of protein conformations through sequence data.

A groundbreaking architecture in this field is AlphaFold2 (AF2) [32], which directly predicts the 3D structures of proteins based on their sequences and evolutionary information. The success of AF2 primarily comes from its innovative architectural designs, particularly the Evoformer and Structure modules. The former extracts and integrates evolutionary information from multiple sequence alignments (MSAs), while the Structure module converts this information into a 3D structure, achieving highly accurate predictions for individual proteins. For complex structure prediction, incorporating evolutionary information from multiple sequences is crucial. This requires aligning related protein sequences and extracting information on co-evolved residues from paired MSAs. To address this, Deep Mind developed AlphaFold-Multimer (AFM) [33] to extract paired MSAs for complex structure prediction. However, when there are too many input sequences or when predicting the structures of orphan proteins, paired MSA information is often insufficient [153]. Since a single genome may contain multiple paralogs of the target sequence, the co-evolutionary signals between different paralogs may interfere with each other, leading to the capture of incorrect co-evolutionary signals in the MSA. This complicates the pairing of different protein chain sequences, making the achievement of paired MSAs for complexes a long-standing challenge and bottleneck [154].

To address the challenges posed by paired MSAs, it is necessary to avoid multi-sequence inputs and adopt several strategies to indirectly predict protein complex structures, leading to the development of some specialized methods. For example, AF2-linker [155, 156] simplifies the multi-sequence prediction task to a single sequence by linking two protein chains with a polypeptide linker. Similarly, AF2-gap [157] effectively handles gaps or missing regions in protein sequences by merging two protein sequences into one and adding position gaps to distinguish different proteins. AF2Complex [158] adopts a similar strategy to AF2-gap and uses gap-filling techniques to extend each monomer MSA to match the complex's length.

Beyond the well-known AF2 and AFM, the same team further developed the specialized AlphaFold3 [34] (AF3) that has the capability to predict the joint complexes for almost all biomolecular systems. AF3 further simplifies the process by reducing the number of MSA modules and substituting the Evoformer with a more efficient Pairformer, thereby reducing dependency on MSA data and emphasizing pair information. Additionally, the structure module of AF3 incorporates a diffusion-based

framework to directly predict atomic coordinates, markedly enhancing efficiency for protein complex prediction.

With the advancement of protein language models (PLMs), new MSA-based methods [159, 160] that can be integrated with architectures such as AF2 have been developed for high-quality protein complex predictions. Chen et al. proposed ESM-Pair [161], which leverages column-wise attention scores from ESM (Evolutionary Scale Modeling)-MSA-1b to identify and match homologs from single-chain MSAs, ultimately enhancing prediction accuracy when input into AF2. Mirdita et al. presented ColabFold [162], a fast and user-friendly tool that enhances sequence search in MMseqs2 for AF2, enabling both homomeric and heteromeric complex modeling for efficient protein complex prediction. Baker et al. introduced RoseTTAFold [163], an end-to-end tri-track network based on paired sequence alignments to predict protein complexes. Zhang et al. introduced DeepMSA [164], a streamlined pipeline designed to generate high-quality MSAs for challenging sequences, later refining the approach to develop DeepMSA2 [165], which was then combined with AFM to create the DMFold-Multimer model for predicting protein complexes.

Alternatively, some approaches bypass MSA entirely for complex structure prediction. For instance, Zhu et al. developed Uni-Fold MuSe [166], a sophisticated structure predictor that uses only amino acid sequences derived from pretrained protein language models. This model relies on homolog search and pairing steps but exclusively uses pretrained PLMs to capture evolutionary patterns, eliminating the need for MSAs.

3.2.2 | Multi-Step Integration Methods

Although sequence-based methods can generate the final complex structure without any traditional docking operation, they are still constrained by the availability of paired MSAs. Many proteins lack homologs, and for these proteins, approaches like the early AF2-linker approach can be employed, where PLMs predict individual protein structures and link them into a single chain [167, 168]. However, this special treatment does not fully address the issue, thus leading to the development of strategies that combine sequence-based approaches with traditional docking methods in a multi-step procedure for complex structure prediction.

Traditional docking methods based on sampling and scoring can finely handle PPIs, but their performance is highly dependent on the quality of initial structures. Conversely, sequence-based structural prediction methods can generate high-quality monomer protein structures but often struggle with accurately defining PPI interfaces. Therefore, combining these approaches into a multi-step framework maximizes their respective strengths, offering a more robust protein docking prediction, particularly for novel and flexible protein complexes.

Ghani et al. proposed a multi-step strategy that integrates ClusPro with AF2 [169]. ClusPro utilizes FFT for sampling and energy-based scoring functions to evaluate PPIs, which can be further enhanced through AF2 to improve docking accuracy. Initially, a strategy similar to AF2-gap is employed, where gap-filling is used to input target protein sequences into AF2, generating initial protein complex structures along with

predicted aligned error (PAE) values. Subsequently, monomer structures are extracted from these complexes and re-docked using ClusPro to generate multiple possible docking conformations. The PAE information from AF2 guides the selection of these conformations, optimizing the docking interface details. Finally, the scores from both AF2 and ClusPro are combined to rank all docking results, with the top-ranked solution chosen as the final prediction.

Similarly, Harmalkar et al. proposed AlphaRED [170], a pipeline that uses AFM as a structure template generator, and ReplicaDock2.0 for subsequent docking. ReplicaDock2.0, a physics-based docking protocol, has the capability to conduct both the exhaustive docking simulation and local flexible docking. Initially, protein complex structures are predicted using AFM implemented in ColabFold. Then, an exhaustive docking simulation is conducted to sample potential docking interfaces, followed by local flexible docking based on the per-residue predicted local distance difference test scores (pLDDT) from AFM. This process optimizes candidate complex structures through backbone movement and flexible adjustments. Finally, the Rosetta tool refines these structures, thus selecting the highest-scoring model.

Multi-step prediction approaches are also applicable to multimetric structures with more than three chains, where direct complex prediction tools like AFM often face accuracy challenges and GPU memory limitations. MolPC [171] proposed by Bryant et al. relies on AF2 to produce configurations for pairs and triplets of chains, and then assembles them using Monte Carlo Tree Search from high-quality predicted subcomponents. CombFold [172] proposed by Shor et al. further considers the complexity of interactions among multiple subunits in large protein complexes. It uses AF2 to generate a limited number of pairwise subunit interactions for assembly and then exhaustively enumerates potential assembly trees during the hierarchical and combinatorial assembly stage, thereby maximizing the probability of accurately assembling the complex based on pairwise AF2 interactions.

3.2.3 | Multimodal Fusion Architectures

Multi-step integration methods in protein docking still have several disadvantages. First, these methods often depend on multiple intermediate steps, which introduce a series of assumptions and potential errors. The accumulation of these errors across multiple steps can lead to substantial biases. Additionally, the integration of different techniques may complicate parameter optimization and compatibility, and the incorporation of traditional algorithms can reintroduce extensive sampling, making the process excessively complex and time-consuming.

The multimodal fusion architecture integrates diverse data sources, including structural information and sequence data. These data are first standardized and cleaned to ensure consistent quality, and then fed into neural networks such as PLM and GNN to extract features at both the sequence and structure levels. These features are subsequently fused using attention mechanisms or interaction layers, enabling the interplay of the characteristics of the two modalities. Finally, the combined

sequence and structure information is used to generate the final docking conformation. Compared to multi-step methods, the multimodal fusion architecture reduces the cumulation of errors and is more suitable for handling protein flexibility.

Luo et al. proposed xTrimoDock [173], a model that integrates protein sequence and single-chain structural information by modifying the Evoformer module to handle and integrate multimodal data. Through multiple layers of transformations and attention mechanisms, xTrimoDock captures the complex relationships between sequences and structures, representing them as predicted inter-protein distance maps. The model then minimizes the difference between the predicted and true distance maps to learn rotational and translational transformations, which could be considered a non-convex optimization process, ultimately solved using spectral initialization and gradient descent. Additionally, the approach employs a bi-level optimization strategy to optimize both the transformations and the hyperparameters of the cross-modal transformer, further enhancing model performance and making it applicable to antigen-antibody docking tasks [174]. However, xTrimoDock emphasizes the overall rotation and translation transformations of proteins, neglecting the fluctuations of individual residues during binding.

To better consider protein flexibility, McPartlon et al. proposed DockGPT [30], which constructs a matrix containing distance and orientation information between residues, thereby transforming the complex problem of protein flexible docking into a matrix completion problem. During training, the model learns to reconstruct complete geometric relationships from partial data using a vast number of samples with known complete matrices. In practical application, DockGPT extracts residue coordinates and orientations from protein structures and feature embeddings from sequences, constructing them in matrix form. Then, a neural network completes the matrix, which is subsequently transformed into a 3D geometric structure to restore the binding state of protein complexes, circumventing the computationally intensive challenge associated with flexible sampling. Additionally, DockGPT utilizes the Rosetta FastRelax protocol to generate simulation data of unbound states, serving as a reference for flexibility optimization. Similarly, Chu et al. developed GeoDock [31], another tool that supports flexible protein-protein docking. The approach first uses ESM-2 to embed sequence features, thus avoiding MSA during training, and then multimodal features of unbound structures are fed into an equivariant attention module to realize protein backbone movement. Both DockGPT and GeoDock can choose to incorporate binding site information and demonstrate impressive flexible docking performance when binding sites are well defined. However, they both treat the docking problem as a regression task, typically providing only a single conformation. As a result, they exhibit relatively lower success rates in blind docking, indicating that such DL models still face challenges in site identification and capturing conformational changes within backbone networks.

4 | Datasets

The success of DL algorithms relies heavily on the quality and scale of the datasets. Therefore, high-quality and extensive datasets are critical for DL models applied to protein-protein

docking. Recent advancements in high-throughput X-ray crystallography and cryo-EM have significantly expanded the number of high-resolution structures available in the PDB, providing a rich resource for these data-driven models. This section offers an overview of some datasets commonly used for training and evaluating DL models in protein–protein docking, as summarized in Table 3.

4.1 | Protein Docking Benchmarks

Docking benchmarks (DB) [175–179] are a set of benchmark datasets renowned for their high-quality data manually curated by domain experts. These datasets provide both unbound–unbound and unbound–bound test cases to facilitate the development and comparison of docking algorithms. The earliest version includes 59 test cases categorized into enzyme–inhibitor complexes, antibody–antigen complexes,

other complexes, and difficult test cases. Starting from DB2 [176], the authors refined the standards for protein selection and redundancy elimination, which improved the quality of the selected protein complexes. The dataset is further divided into three subsets: rigid-body cases, medium-difficulty cases, and high-difficulty cases. Subsequent updates, including DB3 [177], DB4 [178], and DB5.5 [179], use similar standards in dataset curation and have become the gold standard for validating DL models. The latest DB5.5 has collected a total of 257 protein binary complexes.

SDBenchmark [180] is a benchmark dataset specifically designed for evaluating symmetric protein docking, with a primary focus on homologous oligomers with specific symmetries. The dataset covers proteins with various symmetry types, such as cyclic symmetry, dihedral symmetry, cubic symmetry, and helical symmetry, which facilitate a comprehensive assessment of symmetrical protein docking.

TABLE 3 | Summary of some commonly-used datasets.

Dataset	Contained complexes	Website
Docking Benchmark [175]	59 (22 enzyme–inhibitor complexes, 19 antibody–antigen complexes, 11 other complexes, 7 difficult test cases)	https://zlab.wenglab.org/benchmark/
Docking Benchmark2 [176]	72 (52 rigid-body cases, 13 medium-difficulty cases, 7 high-difficulty cases)	
Docking Benchmark3 [177]	124 (88 rigid-body cases, 19 medium-difficulty cases, 17 high-difficulty cases)	
Docking Benchmark4 [178]	176 (121 rigid-body cases, 30 medium-difficulty cases, 25 high-difficulty cases)	
Docking Benchmark5.5 [179]	257 (162 rigid-body cases, 60 medium-difficulty cases, 35 high-difficulty cases)	
DOCKGROUND [193, 194]	667,331 (bound set update weekly); 396 X-ray unbound and 3205 simulated unbound complexes (unbound set); 171 binary complexes and 963 modeled complexes (model-model complexes); 457 targets, 99 nonnative and 1 near-native match (Docking decoys); 12,470 complexes (Docking templates)	https://dockground.compbio.ku.edu/
DIPS [183]	42,826	https://github.com/drorlab/DIPS
DIPS-Plus [184]	42,112	https://github.com/BioinfoMac hineLearning/DIPS-Plus
SAbDab [190, 191]	5426 (update weekly)	http://opig.stats.ox.ac.uk/webapps/sabdab
RAbD [192]	60	http://dunbrack2.fccc.edu/PyIgClassify/default.aspx
PDBbind2021 [186]	3176	https://www.pdbbind-plus.org/cn/
PINDER [196]	2,319,564 (dimeric complex)	https://github.com/pinder-org/pinder
PPI4Dock [195]	1417	http://biodev.cea.fr/inter evol/ppi4dock/
SDBenchmark [180]	251	http://huanglab.phys.hust.edu.cn/SDBenchmark/

The Critical Assessment of Predicted Interactions (CAPRI) challenge is a community experiment aimed at assessing protein–protein docking prediction methods [181, 182]. Launched in 2001, CAPRI has encouraged researchers to conduct blind docking predictions by providing unpublished protein complex structures as docking targets. The goal of CAPRI is to evaluate the performance of docking algorithms by comparing predicted models with actual structures, particularly with regard to predicting interaction regions and residue contacts. To date, CAPRI has completed 57 assessment rounds, driving improvements in protein–protein docking algorithms and providing deeper insights into PPIs.

4.2 | Datasets With Interaction Information

The Database of Interacting Protein Structures (DIPS) [183] is built by mining PPI pairs directly from the PDB. The retrieved structures should satisfy the following criteria: they must have a buried surface area $\geq 500 \text{ \AA}^2$, be determined by X-ray crystallography or cryo-EM with the resolution $\leq 3.5 \text{ \AA}$, contain only protein chains longer than 50 amino acids, and represent proteins in the bound state. To avoid possible duplications with DB5.5, sequence-based pruning is applied to exclude complexes with more than 30% sequence identity to any protein in DB5.5. Additionally, DIPS even provides the interacting residues for each PPI pair, defined as amino acids within $6 \text{ \AA}$ of any non-hydrogen atoms from the paired protein. DIPS contains 42,826 binary complexes, far larger than DB5.5, and is currently widely used for training DL models.

DIPS-Plus [184] enhances the DIPS dataset by adding new residue-level features while maintaining similar selection criteria. These added features include surface proximity, half-sphere amino acid composition, and profile hidden Markov model (HMM)-based sequence features, which enrich the dataset with extensive protein information [185]. This makes DIPS-Plus particularly useful for calculating and predicting various downstream protein-related tasks. Additionally, DIPS-Plus provides a feature extraction script to facilitate the conversion of protein features into graph representations, enabling the application of geometric learning algorithms.

The PDBbind dataset [186] is a curated, high-resolution subset of PDB, updated to PDBbind2021, which includes thousands of protein complexes and their binding affinity data (K_d , K_i , IC_{50}), which is invaluable for modeling DL-based models related to PPIs. As such, the dataset serves as a valuable supplement to DL-based docking and interaction models.

4.3 | Antibody-Related Datasets

Antibody drugs exert therapeutic effects through binding to specific antigens. Protein–protein docking technology can simulate their binding, thereby facilitating the design and optimization of antibody binding sites [187–189]. Consequently, antibody structural data are often used as an extra resource to validate protein–protein docking tools to simulate more realistic drug design scenarios.

Structural Antibody Database (SAbDab) [190, 191] is a weekly-updated database consisting of thousands of antibody–antigen complex structures. It employs the Chothia numbering scheme to automatically annotate antibody chains, distinguish heavy and light chains, and pair them accordingly. The database provides detailed annotations of the antibody's complementarity-determining regions (CDRs) and framework regions (FRs), along with details on antibody–antigen binding sites and binding affinity. These annotations help users better understand antibody structure and function, facilitating structural comparisons and functional predictions.

RABD [192] is a benchmark dataset for antigen–antibody complexes, widely used in antibody design evaluations. To avoid bias from training set overlap, some studies also explore relatively new antibody and antigen data from the PDB, such as antibodies (VH-VL) and antibody–antigen (AB-AG) data [173, 174]. These datasets allow for separate evaluations of variable heavy-light chains and antigen–antibody docking, making them useful for studying the role of protein–protein docking in antibody research.

4.4 | Datasets With Simulated Complex Predictions

The DOCKGROUND database [193, 194] is a comprehensive resource that provides five interrelated categories for different stages of protein–protein docking development and testing: bound dataset, unbound datasets, model–model complexes, docking decoys, and docking templates. The bound and unbound datasets form the core of the resource, which is derived from experimentally determined structures. The remaining categories, such as model–model complexes and docking decoys, are generated through protein structure modeling predictions with varying degrees of accuracy. Additionally, the decoy predictions can be used for the development and validation of scoring functions. A key feature of DOCKGROUND is its user-friendly web interface, allowing researchers to conveniently download precompiled or customized datasets based on their specific requirements.

The PPI4DOCK dataset [195] is a large-scale collection for evaluating the role of homology modeling in docking predictions. It includes both experimentally determined protein structures and homology-modeled protein structures, allowing for a direct comparison of docking performance using high-quality experimental data versus predicted structures. This makes PPI4DOCK particularly valuable for assessing the accuracy and applicability of homology modeling-based docking methods, especially when experimental structures are unavailable or nonexistent. However, it is important to note that, since the structures are derived from unbound templates via homology modeling, their evaluation may be influenced by modeling inaccuracies, and, as such, PPI4DOCK cannot be considered a strict benchmark.

The PINDER dataset [196] is a highly annotated resource that uses structural clustering to derive data partitions based on non-redundant interfaces, including holo, apo, and predicted structures. Its diversity is reflected in the broad range of metadata

annotations, including residue and chain lengths, buried surface areas, and sectional properties. Furthermore, PINDER offers custom dataset interfaces like ProteinFlow [197], which use innovative splitting algorithms combined with similarity metrics like Foldseek and MM-seq2 to ensure diversity in test and validation sets.

5 | Evaluation

In protein–protein docking tasks, another important aspect is evaluating the final docking results, which primarily involves two aspects: assessing the quality of the binding modes obtained after protein docking and quantifying the performance of docking algorithms based on the quality of these binding modes. The former involves comparing predicted binding modes with experimentally determined native complex structures to validate prediction accuracy, while the latter evaluates the overall performance of different algorithms under the same conditions and datasets, allowing for a direct comparison of their performance and applicability.

5.1 | Binding Mode Evaluation

The CAPRI criteria are the most widely recognized for evaluating the quality of predicted binding modes, primarily based on a combination of three criteria [181, 182]: (i) the root mean square deviation of the ligand protein (L-RMSD), which measures the global geometric fit of the predicted conformation by calculating the RMSD of the ligand protein after optimal superposition with the native receptor protein; (ii) the interface root mean square deviation (I-RMSD), which quantifies the fit of the interface region by calculating the RMSD of main-chain atoms between the native and predicted interfaces after optimal superposition; and (iii) the fraction of native contacts (f_{nat}) in the predicted complex, which represents the quality of predicted interactions on the interface by calculating the percentage of native residue–residue contacts in the predicted interface. According to these criteria, the CAPRI standards classify the accuracy of predictions into four standards: high, medium, acceptable, and incorrect. Predictions in incorrect categories are considered random solutions, while the others only consider the top-ranked hits.

Individual indicators in the CAPRI criteria can also serve as evaluation standards. L-RMSD, which requires aligning only the receptor protein without considering the overlap between the native and predicted binding interfaces, commonly uses a threshold of $\text{L-RMSD} \leq 10 \text{ \AA}$ to determine success [50, 198]. In contrast, I-RMSD focuses on interface agreement, with a typical success threshold of $\text{I-RMSD} \leq 2.5 \text{ \AA}$ [41, 49]. A new indicator, C-RMSD [24], represents the root mean square deviation of the entire protein complex. It superimposes the predicted complex structure onto the native complex structure using the Kabsch algorithm and calculates the C-RMSD, providing a more comprehensive metric that accounts for changes in both the ligand and receptor proteins. This makes C-RMSD particularly suitable for evaluating DL-based approaches. The use of individual RMSD indicators can effectively reduce computational costs and provide a more comprehensive assessment of the performance for specific tasks through the adjustment of different RMSD cutoffs.

5.2 | Docking Algorithm Evaluation

CAPRI criteria can only estimate the quality of a binding pose and do not directly reflect the performance of a docking algorithm. Therefore, there is a need to design a single robust continuous quality estimate method to cover all the diverse structural attributes individually assessed by the CAPRI measures, thus deriving a variety of indicators, such as $\text{C}\alpha$ -RMSD, TM-score [199], GDT-TS [200], IS-Score [201], IDDT [202], DockQ [203], and pDockQ [157]. These indicators are somewhat correlated, but they emphasize different aspects of the prediction results. Taking DockQ as an example, it incorporates three CAPRI metrics and is formulated as a continuous variable:

$$\text{RMS}_{\text{scaled}}(\text{RMS}, d_i) = \frac{1}{1 + \left(\frac{\text{RMS}}{d_i}\right)^2} \quad (16)$$

$$\text{DockQ} = \frac{F_{\text{nat}} + \text{RMS}_{\text{scaled}}(\text{LRMS}, d_1) + \text{RMS}_{\text{scaled}}(\text{IRMS}, d_2)}{3} \quad (17)$$

where *DockQ* is the final score, $\text{RMS}_{\text{scaled}}$ normalizes RMSD by incorporating both L-RMSD and I-RMSD and fixing them within a specific range, and d_i is a scaling factor. The essence of DockQ lies in its integration of three crucial metrics from the CAPRI criteria, essentially encompassing CAPRI classification to provide a more nuanced assessment. Such scoring systems integrate multiple quality indicators, providing a continuous and interpretable score, which is more suitable for evaluating and comparing the accuracy of docking algorithms. Table 4 summarizes various evaluation metrics and the meanings associated with their different parameter ranges.

In addition to accuracy assessment, another performance comparison for docking algorithms is based on success rate and hit count, similar to methods used in small molecule docking tasks [39]. The success rate refers to the percentage of correctly predicted conformations among a given number of targets in the test set. When calculating the success rate, the top-ranking binding modes are usually considered, typically using the top 10 binding modes for protein docking tasks. In other words, if any of the top 10 binding modes hits the target, it is considered a successful prediction of the native structure. A higher success rate reflects stronger predictive capabilities of the docking algorithm across different targets, making it suitable for a wider range of applications. The hit count refers to the number of correctly predicted conformations among a set of top-scoring docking poses. Hit count is similar to enrichment factors in the protein–small molecule docking field, often calculated based on the top N conformations for a single target. A higher hit count directly reflects the algorithm's ability to generate high-quality initial binding conformations, which is beneficial for subsequent selection and optimization processes.

6 | Concluding Remarks and Future Perspectives

The introduction of advanced DL techniques into the field of protein–protein docking has posed a tremendous impact on protein research. In this review, we systematically examine approaches related to protein docking. First, we provide a concise overview

TABLE 4 | Summary of some evaluation metrics and the meanings associated with their different parameter ranges.

Accuracy category	Task and criteria
<i>CAPRI criteria</i> [181, 182]	<i>Evaluate the quality of protein docking</i>
High	$f_{\text{nat}} \geq 0.5$ and ($L_{\text{rmsd}} \leq 1.0$ or $I_{\text{rmsd}} \leq 1.0$)
Medium	$f_{\text{nat}} \geq 0.3$ and ($1.0 < L_{\text{rmsd}} \leq 5.0$ or $1.0 < I_{\text{rmsd}} \leq 2.0$)
Acceptable	$f_{\text{nat}} \geq 0.1$ and ($5.0 < L_{\text{rmsd}} \leq 10.0$ or $2.0 < I_{\text{rmsd}} \leq 4.0$)
Incorrect	$f_{\text{nat}} < 0.1$ or ($L_{\text{rmsd}} > 10.0$ and $I_{\text{rmsd}} > 4.0$)
<i>IS-score</i> [201]	<i>Evaluate the quality of protein docking</i>
High	ISscore ≥ 0.69
Medium	$0.48 \leq \text{ISscore} < 0.69$
Acceptable	$0.26 \leq \text{ISscore} < 0.48$
Incorrect	ISscore < 0.26
<i>DockQ</i> [203]	<i>Evaluate the quality of protein docking</i>
High	$\text{DockQ} \geq 0.80$
Medium	$0.49 \leq \text{DockQ} < 0.80$
Acceptable	$0.23 \leq \text{DockQ} < 0.49$
Incorrect	$\text{DockQ} < 0.23$
<i>L-RMSD</i>	<i>Evaluate the quality of protein docking</i>
Successful	$L_{\text{rmsd}} \leq 10.0$
Incorrect	Otherwise
<i>I-RMSD</i>	<i>Evaluate the quality of protein docking</i>
Successful	$I_{\text{rmsd}} \leq 2.5$
Incorrect	Otherwise
<i>TM-score</i> [199]	<i>Evaluate the structural similarity of proteins</i>
Highly relevant	TMscore ≥ 0.50
Potentially relevant	$0.17 \leq \text{TMscore} < 0.50$
Irrelevant	TMscore < 0.17
<i>GDT-TS or GDT-HA</i> [200]	<i>Evaluate the structural similarity of proteins (global distance differences)</i>
The higher, the better	$\text{GDTTS} \in (0, 1), \text{GDTHA} \in (0, 1)^a$
<i>lDDT</i> [202]	<i>Evaluate the structural similarity of proteins (local distance differences)</i>
The higher, the better	$\text{lDDT} \in (0, 1)$

^aGDT-TS and GDT-HA only differ in accuracy. The former is of low accuracy and the latter is of high accuracy.

of various traditional docking strategies. Subsequently, we categorize DL-based methods into two groups: rigid-body docking and flexible docking. For the former, structure-based strategies have gained significant attention, with emerging approaches such as GDL, EGNN, and DM as notable examples. For the latter, sequence-based strategies are receiving increasing focus, particularly in the context of protein complex co-folding. Finally, we summarize the commonly used datasets and evaluation metrics in the protein–protein docking field.

Figure 4 illustrates several tasks derived from protein docking. Multimer docking involves assembling multiple protein subunits into a protein complex. Compared to dimer docking, multimer docking requires multiple pairwise predictions while accounting for interactions and spatial arrangements among multiple proteins. Although traditional methods like HADDOCK and DL-based approaches such as AFM have achieved notable progress, they remain constrained by the speed of sampling and MSAs. Therefore, further advancements in the efficiency of DL models are necessary. Symmetry protein docking focuses on modeling interactions within symmetric protein complexes, particularly in homodimers, trimers, or higher-order oligomers [215]. A substantial number of experimentally resolved proteins exhibit certain types of symmetry, making symmetry docking an important area worthy of study. Unlike direct sampling-based docking, symmetry protein docking emphasizes strict symmetry constraints and accommodates various symmetry modes, including cyclic (C_n) symmetry, polyhedral (D_n) symmetry, and helical symmetry [216]. Although several models [143, 150, 171] have been developed to address this challenge, a unified benchmark is still needed to comprehensively evaluate these methods. Protein flexibility represents a core challenge in DL-based protein docking that urgently needs to be addressed. Docking starting from apoprotein structures still fails to yield satisfactory results. From the perspective of DL, the flexibility problem involves updating the coordinates of individual residues, necessitating more suitable model frameworks for global optimization and refinement of the entire proteins. Moreover, the issue of flexibility extends to the current state of intrinsically disordered protein (IDP) docking. IDPs lack stable or well-defined tertiary structures under physiological conditions, often containing disordered regions with no clear secondary structure [217, 218]. This makes experimental structural analysis of IDPs highly challenging, resulting in the shortage of structural data, which significantly hampers the progress of DL-driven approaches. Unlike globular proteins, IDPs are characterized by disordered regions that can undergo large conformational changes, adding an extra layer of complexity to the docking process. Therefore, DL models must account for the dynamic behavior and conformational heterogeneity of IDPs, enabling them to explore the broad conformational space. Overcoming these limitations may require the integration of techniques such as MD simulations, as well as the development of IDP-specific datasets that capture various conformational states for DL model training. Research on antigen–antibody docking has garnered significant attention due to its importance in immunotherapy and vaccine development. While substantial progress has been made in designing antibodies targeting variable regions, challenges remain in recognizing diverse antigen epitopes and predicting docking poses. Direct application of DL-based protein docking tools to antigen–antibody complexes is

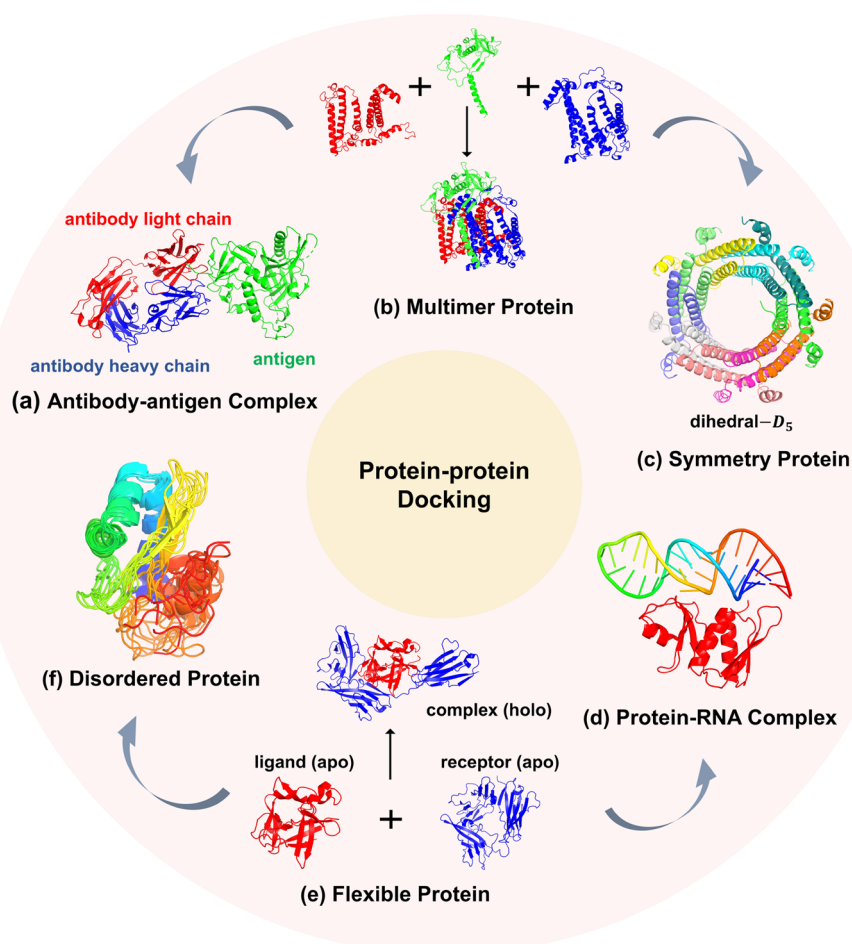


FIGURE 4 | Derived targets and tasks in protein-protein docking. (a) An example of an antigen-antibody complex structure (PDB ID: 6MG7), comprising the antibody heavy chain (blue), antibody light chain (red), and antigen (green); (b) Docking of a multimeric complex (PDB ID: 7MH3); (c) An example of a symmetry protein structure (PDB ID: 7S8T), exhibiting dihedral- D_5 symmetry; (d) An example of an RNA-protein complex structure (PDB ID: 1I6U), with the protein shown in red and the RNA in multiple colors; (e) Docking of flexible proteins (PDB ID: 1IRA), where both the receptor and ligand are in apo states, resulting in significant structural changes in the receptor post-docking; (f) An example of an intrinsically disordered protein structure (PDB ID: 8TB1), with disordered regions highlighted in red and orange.

feasible, but their accuracy is notably lower [219]. This is largely due to the lack of co-evolutionary information between antigens and antibodies, leading to suboptimal performance of frameworks like AF2. Thus, developing antigen-antibody docking methods must emphasize geometric and chemical information as critical factors. Nucleic acid-protein complexes represent a relatively novel docking target. However, the shortage of experimentally validated structures and the dynamic nature of nucleic acid complicate computational modeling [220]. Addressing these challenges requires specialized docking algorithms that account for the structural flexibility of nucleic acid.

DL-based protein docking methods have seen a surge in development. While various models successfully enable docking and significantly surpass traditional methods in speed by bypassing direct sampling processes, their docking accuracy still falls short compared to conventional approaches. In competitions such as CAPRI and CASP [221], aside from the notable performance of protein-folding models based on the AF2 framework, other structure-based DL methods struggle to compete with traditional docking algorithms. One key advantage of the DL strategies based on the AF2 framework is their ability to effectively

leverage MSAs. Through MSAs, DL models can learn sequence variations and conservation across different species, which are then incorporated into 3D structure modeling to improve structural predictions. In the docking field, the homologous information provided by MSAs helps the model identify potential binding interfaces and optimize docking accuracy. In contrast, other DL methods based on GNN or DM architectures typically use only structural information from limited datasets, focusing on contact maps or physical features without specifically designing for evolutionary information. These models may fail to fully understand the conservation and variability of protein sequences and struggle to capture the evolutionary relationships between sequences, which could be a key reason for the underperformance of structure-based DL strategies.

Moreover, several issues persist in the docking results of different models. For instance, ligand conformations predicted by EGNN-based docking often overlap significantly with receptor proteins, suggesting a need to introduce penalties for atomic clashes caused by excessively close interatomic distances. Similarly, ligand conformations derived from DM-based docking sometimes maintain a certain distance from receptor

proteins, failing to achieve a true binding state, indicating the necessity for models to enforce reasonable sampling ranges. Multimodal end-to-end generative models for predicting protein complexes exhibit noticeable performance degradation in the absence of binding site information, and the generated flexible conformations are often unreasonable, further emphasizing the importance of binding context. In addition, the evaluation standards for current DL-based protein docking models are problematic. Ganea et al. were among the first to evaluate DL-based models, employing the DIPS split and DB5 training-testing set partitioning as mainstream benchmarks [24]. However, reports have indicated that the DIPS dataset split exhibits a high degree of similarity between the training and test sets, potentially leading to inflated test results [149]. Consequently, subsequent studies have adopted alternative training and testing standards, making direct comparisons between models challenging. Therefore, the DL-based protein docking field urgently requires a unified and suitable benchmark, aside from the CAPRI and CASP competitions, to enable consistent evaluation of these models.

However, structure-based DL methods still exhibit significant potential. These approaches generally demonstrate lower computational costs and faster inference speeds, rendering them more efficient for large-scale complex docking tasks. The heavy reliance of the AF2 framework on MSAs may limit its predictive capabilities for orphan proteins with scarce evolutionary information [153]. In contrast, structure-based methods can achieve precise predictions for such orphan proteins by incorporating transfer learning [222], few-shot learning [223], or integrating physical prior knowledge [224], thereby enhancing the overall robustness and generalization capability of the models. Future advancements in DL-driven docking methodologies may focus on the following pivotal directions: first, exploring cross-modal learning by integrating sequence, structure, and evolutionary information to enhance the effective use of MSAs, ensuring competitiveness even when MSA quality is low, and improving adaptability and generalization for complex docking tasks; second, for tasks such as orphan protein docking, leveraging physics-based information and fine-tuning with small sample data to optimize docking performance for specialized scenarios, facilitating drug discovery and design in unique contexts.

Beyond the discussions above, the future of protein docking lies in the development of novel and effective end-to-end DL-based docking algorithms. Although DL-driven molecular docking has made significant progress in recent years, DL-based protein docking has only recently gained attention and development, primarily relying on protein folding frameworks. With the increasing availability of high-quality protein complex datasets, DL-based protein docking is expected to become mainstream. This review highlights the importance of protein docking algorithms by comprehensively comparing various computational architectures, algorithm designs, and evaluation metrics, elucidating their strengths and limitations. These insights aim to guide the development of more efficient docking algorithms. Undoubtedly, the continuous improvement of DL-based protein docking methods will accelerate the identification of new PPIs and drugs. We hope this review offers valuable perspectives for future research in this field.

Author Contributions

Linlong Jiang: formal analysis (lead), investigation (lead), methodology (lead), visualization (lead), writing – original draft (lead). **Ke Zhang:** conceptualization (equal), investigation (equal), writing – original draft (lead). **Kai Zhu:** investigation (supporting), visualization (supporting), writing – original draft (supporting). **Hui Zhang:** investigation (supporting), visualization (supporting), writing – original draft (supporting). **Chao Shen:** conceptualization (equal), project administration (equal), supervision (equal), writing – original draft (equal), writing – review and editing (supporting). **Tingjun Hou:** conceptualization (lead), funding acquisition (lead), project administration (lead), supervision (lead), writing – review and editing (lead).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Related WIREs Articles

[Computational prediction of protein-protein binding affinities](#)

[Structural prediction of protein interactions and docking using conservation and coevolution](#)

References

1. E. M. Phizicky and S. Fields, “Protein-Protein Interactions: Methods for Detection and Analysis,” *Microbiological Reviews* 59 (1995): 94–123.
2. M. R. Arkin and J. A. Wells, “Small-Molecule Inhibitors of Protein-Protein Interactions: Progressing Towards the Dream,” *Nature Reviews Drug Discovery* 3 (2004): 301–317.
3. M. G. Kann, “Protein Interactions and Disease: Computational Approaches to Uncover the Etiology of Diseases,” *Briefings in Bioinformatics* 8 (2007): 333–346.
4. J. C. Fuller, N. J. Burgoyne, and R. M. Jackson, “Predicting Druggable Binding Sites at the Protein-Protein Interface,” *Drug Discovery Today* 14 (2009): 155–161.
5. S. Grosdidier and J. Fernandez-Recio, “Protein-Protein Docking and Hot-Spot Prediction for Drug Discovery,” *Current Pharmaceutical Design* 18 (2012): 4607–4618.
6. I. A. Vakser, “Protein-Protein Docking: From Interaction to Interactome,” *Biophysical Journal* 107 (2014): 1785–1793.
7. M. Zhou, Q. Li, and R. Wang, “Current Experimental Methods for Characterizing Protein-Protein Interactions,” *ChemMedChem* 11 (2016): 738–756.
8. O. Keskin, A. Gursoy, B. Ma, and R. Nussinov, “Principles of Protein-Protein Interactions: What Are the Preferred Ways for Proteins to Interact?,” *Chemical Reviews* 108 (2008): 1225–1244.
9. G. R. Smith and M. J. Sternberg, “Prediction of Protein-Protein Interactions by Docking Methods,” *Current Opinion in Structural Biology* 12 (2002): 28–35.
10. S.-Y. Huang and X. Zou, “Advances and Challenges in Protein-Ligand Docking,” *International Journal of Molecular Sciences* 11 (2010): 3016–3034.
11. D. W. Ritchie, “Recent Progress and Future Directions in Protein-Protein Docking,” *Current Protein and Peptide Science* 9 (2008): 1–15.

12. J. Biesiada, A. Porollo, P. Velayutham, et al., "Survey of Public Domain Software for Docking Simulations and Virtual Screening," *Human Genomics* 5 (2011): 1–9.
13. I. H. Moal, M. Torchala, P. A. Bates, and J. Fernández-Recio, "The Scoring of Poses in Protein-Protein Docking: Current Capabilities and Future Directions," *BMC Bioinformatics* 14, no. 1 (2013): 286, <https://doi.org/10.1186/1471-2105-14-286>.
14. T. Shimoda, S. Suzuki, M. Ohue, et al., "Protein-Protein Docking on Hardware Accelerators: Comparison of GPU and MIC Architectures," *BMC Systems Biology* 9 (2015): 1–10.
15. A. A. Kaczor, *Protein-Protein Docking: Methods and Protocols* (Springer Nature, 2024).
16. Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," *Nature* 521 (2015): 436–444.
17. J. Schmidhuber, "Deep Learning in Neural Networks: An Overview," *Neural Networks* 61 (2015): 85–117.
18. I. Castiglioni, L. Rundo, M. Codari, et al., "AI Applications to Medical Images: From Machine Learning to Deep Learning," *Physica Medica* 83 (2021): 9–24.
19. R. Gupta, D. Srivastava, M. Sahu, S. Tiwari, R. K. Ambasta, and P. Kumar, "Artificial Intelligence to Deep Learning: Machine Intelligence Approach for Drug Discovery," *Molecular Diversity* 25 (2021): 1315–1360.
20. S. Dara, S. Dhamercherla, S. S. Jadav, C. M. Babu, and M. J. Ahsan, "Machine Learning in Drug Discovery: A Review," *Artificial Intelligence Review* 55 (2022): 1947–1999.
21. H. Askr, E. Elgeldawi, H. Aboul Ella, Y. A. M. M. Elshaier, M. M. Gomaa, and A. E. Hassanien, "Deep Learning in Drug Discovery: An Integrative Review and Future Challenges," *Artificial Intelligence Review* 56 (2023): 5975–6037.
22. A. Tropsha, O. Isayev, A. Varnek, G. Schneider, and A. Cherkasov, "Integrating QSAR Modelling and Deep Learning in Drug Discovery: The Emergence of Deep QSAR," *Nature Reviews Drug Discovery* 23 (2024): 141–155.
23. D. B. Catacutan, J. Alexander, A. Arnold, and J. M. Stokes, "Machine Learning in Preclinical Drug Discovery," *Nature Chemical Biology* 20 (2024): 960–973.
24. O.-E. Ganea, X. Huang, C. Bunne, et al., "Independent SE (3)-Equivariant Models for End-to-End Rigid Protein Docking," *arXiv* (2021) Preprint arXiv:2111.07786, <https://doi.org/10.48550/arXiv.2111.07786>.
25. F. Sverrisson, J. Feydy, and J. Southern, "Physics-Informed Deep Neural Network for Rigid-Body Protein Docking," in *MLDD 2022-Machine Learning for Drug Discovery Workshop of ICLR 2022*, (2022).
26. M. A. Ketata, C. Laue, R. Mammadov, et al., "Diffdock-Pp: Rigid Protein-Protein Docking With Diffusion Models," *arXiv* (2023) Preprint arXiv:2304.03889, <https://doi.org/10.48550/arXiv.2304.03889>.
27. J. Han, J. Cen, L. Wu, et al., "A Survey of Geometric Graph Neural Networks: Data Structures, Models and Applications," *arXiv* (2024) Preprint arXiv:2403.00485, <https://doi.org/10.48550/arXiv.2403.00485>.
28. J. Yim, H. Stärk, G. Corso, et al., "Diffusion Models in Protein Structure and Docking," *WIREs Computational Molecular Science* 14 (2024): e1711.
29. A. Harmalkar and J. J. Gray, "Advances to Tackle Backbone Flexibility in Protein Docking," *Current Opinion in Structural Biology* 67 (2021): 178–186.
30. M. McPartlon and J. Xu, "Deep Learning for Flexible and Site-Specific Protein Docking and Design," *bioRxiv* 2023 (2023): 535079.
31. L. Chu, J. A. Ruffolo, A. Harmalkar, and J. J. Gray, "Flexible Protein-Protein Docking With a Multitrack Iterative Transformer," *Protein Science* 33, no. 2 (2024): e4862, <https://doi.org/10.1002/pro.4862>.
32. J. Jumper, R. Evans, A. Pritzel, et al., "Highly Accurate Protein Structure Prediction With AlphaFold," *Nature* 596, no. 7873 (2021): 583–589, <https://doi.org/10.1038/s41586-021-03819-2>.
33. R. Evans, M. O'Neill, A. Pritzel, et al., "Protein Complex Prediction With AlphaFold-Multimer," *bioRxiv* 2021 (2021): 463034.
34. J. Abramson, J. Adler, J. Dunger, et al., "Accurate Structure Prediction of Biomolecular Interactions With AlphaFold 3," *Nature* 630 (2024): 493–500.
35. S.-Y. Huang, "Search Strategies and Evaluation in Protein-Protein Docking: Principles, Advances and Challenges," *Drug Discovery Today* 19 (2014): 1081–1096.
36. S. Sunny and P. B. Jayaraj, "Protein-Protein Docking: Past, Present, and Future," *Protein Journal* 41 (2022): 1–26.
37. M. Zacharias, "Accounting for Conformational Changes During Protein-Protein Docking," *Current Opinion in Structural Biology* 20 (2010): 180–186.
38. J. Liu and R. Wang, "Classification of Current Scoring Functions," *Journal of Chemical Information and Modeling* 55 (2015): 475–482.
39. C. Shen, J. Ding, Z. Wang, et al., "From Machine Learning to Deep Learning: Advances in Scoring Functions for Protein-Ligand Docking," *WIREs Computational Molecular Science* 10 (2020): e1429.
40. D. Morozov, A. Melnikov, V. Shete, et al., "Protein-Protein Docking Using a Tensor Train Black-Box Optimization Method," *arXiv* (2023) Preprint arXiv:2302.03410, <https://doi.org/10.48550/arXiv.2302.03410>.
41. Q. Meng, F. Guo, E. Wang, and J. Tang, "ComDock: A Novel Approach for Protein-Protein Docking With an Efficient Fusing Strategy," *Computers in Biology and Medicine* 167 (2023): 107660.
42. S. Kousaka and T. Ishikawa, "Quantum Chemistry-Based Protein-Protein Docking Without Empirical Parameters," *Journal of Chemical Theory and Computation* 20, no. 12 (2024): 5164–5175, <https://doi.org/10.1021/acs.jctc.4c00531>.
43. E. Katchalski-Katzir, I. Shariv, M. Eisenstein, A. A. Friesem, C. Aflalo, and I. A. Vakser, "Molecular Surface Recognition: Determination of Geometric Fit Between Proteins and Their Ligands by Correlation Techniques," *Proceedings of the National Academy of Sciences of the United States of America* 89 (1992): 2195–2199.
44. H. A. Gabb, R. M. Jackson, and M. J. Sternberg, "Modelling Protein Docking Using Shape Complementarity, Electrostatics and Biochemical Information," *Journal of Molecular Biology* 272 (1997): 106–120.
45. R. Chen, L. Li, and Z. Weng, "ZDOCK: An Initial-Stage Protein-Docking Algorithm," *Proteins* 52 (2003): 80–87.
46. A. Heifetz, E. Katchalski-Katzir, and M. Eisenstein, "Electrostatics in Protein-Protein Docking," *Protein Science* 11 (2002): 571–587.
47. C. L. Bajaj, R. Chowdhury, and V. Siddahanavalli, "F² Dock: Fast Fourier Protein-Protein Docking," *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 8 (2009): 45–58.
48. I. A. Vakser, "Evaluation of GRAMM Low-Resolution Docking Methodology on the Hemagglutinin-Antibody Complex," *Proteins* 29 (1997): 226–230.
49. B. G. Pierce, Y. Hourai, and Z. Weng, "Accelerating Protein Docking in ZDOCK Using an Advanced 3D Convolution Library," *PLoS One* 6 (2011): e24657.
50. D. Kozakov, R. Brenke, S. R. Comeau, and S. Vajda, "PIPER: An FFT-Based Protein Docking Program With Pairwise Potentials," *Proteins* 65 (2006): 392–406.
51. D. Kozakov, D. R. Hall, B. Xia, et al., "The ClusPro Web Server for Protein-Protein Docking," *Nature Protocols* 12 (2017): 255–278.
52. Y. Yan, H. Tao, J. He, and S. Y. Huang, "The HDock Server for Integrated Protein-Protein Docking," *Nature Protocols* 15 (2020): 1829–1852.

53. B. Jiménez-García, C. Pons, and J. Fernández-Recio, "pyDockWEB: A Web Server for Rigid-Body Protein-Protein Docking Using Electrostatics and Desolvation Scoring," *Bioinformatics* 29 (2013): 1698–1699.
54. M. Ohue, Y. Matsuzaki, N. Uchikoga, T. Ishida, and Y. Akiyama, "MEGADOCK: An All-To-All Protein-Protein Interaction Prediction System Using Tertiary Structure Data," *Protein and Peptide Letters* 21 (2014): 766–778.
55. D. Padhorny, A. Kazennov, B. S. Zerbe, et al., "Protein-Protein Docking by Fast Generalized Fourier Transforms on 5D Rotational Manifolds," *Proceedings of the National Academy of Sciences of the United States of America* 113, no. 30 (2016): E4286–E4293, <https://doi.org/10.1073/pnas.1603929113>.
56. E. Ramírez-Aportela, J. R. López-Blanco, and P. Chacón, "FRODOCK 2.0: Fast Protein-Protein Docking Server," *Bioinformatics* 32 (2016): 2386–2388.
57. J. I. Garzon, J. R. López-Blanco, C. Pons, et al., "FRODOCK: A New Approach for Fast Rotational Protein-Protein Docking," *Bioinformatics* 25 (2009): 2544–2551.
58. D. W. Ritchie and G. J. L. Kemp, "Protein Docking Using Spherical Polar Fourier Correlations," *Proteins* 39 (2000): 178–194.
59. D. W. Ritchie and V. Venkatraman, "Ultra-Fast FFT Protein Docking on Graphics Processors," *Bioinformatics* 26 (2010): 2398–2405.
60. F. Jiang and S.-H. Kim, "Soft Docking: Matching of Molecular Surface Cubes," *Journal of Molecular Biology* 219 (1991): 79–102.
61. P. N. Palma, L. Krippahl, J. E. Wampler, and J. J. G. Moura, "BiGGER: A New (Soft) Docking Algorithm for Predicting Protein Interactions," *Proteins* 39 (2000): 372–384.
62. G. Terashi, M. Takeda-Shitaka, K. Kanou, M. Iwade, D. Takaya, and H. Umeyama, "The SKE-DOCK Server and Human Teams Based on a Combined Method of Shape Complementarity and Free Energy Estimation," *Proteins* 69 (2007): 866–872.
63. C. Pons, D. Jiménez-González, C. González-Álvarez, et al., "Cell-Dock: High-Performance Protein-Protein Docking," *Bioinformatics* 28 (2012): 2394–2396.
64. X. Ding, Y. Wu, Y. Wang, J. Z. Vilseck, and C. L. Brooks, III, "Accelerated CDOCK With GPUs, Parallel Simulated Annealing, and Fast Fourier Transforms," *Journal of Chemical Theory and Computation* 16 (2020): 3910–3919.
65. V. Venkatraman, Y. D. Yang, L. Sael, and D. Kihara, "Protein-Protein Docking Using Region-Based 3D Zernike Descriptors," *BMC Bioinformatics* 10, no. 1 (2009): 1–21, <https://doi.org/10.1186/1471-2105-10-407>.
66. D. Duhovny, R. Nussinov, and H. J. Wolfson, "Efficient Unbound Docking of Rigid Molecules," in *Proceedings of the Algorithms in Bioinformatics: Second International Workshop, WABI 2002 Rome, Italy*, (Springer, 2002), 185–200.
67. B. K. Shoichet and I. D. Kuntz, "Protein Docking and Complementarity," *Journal of Molecular Biology* 221 (1991): 327–346.
68. B. K. Shoichet and I. D. Kuntz, "Predicting the Structure of Protein Complexes: A Step in the Right Direction," *Chemistry & Biology* 3 (1996): 151–156.
69. D. M. Lorber, M. K. Udo, and B. K. Shoichet, "Protein-Protein Docking With Multiple Residue Conformations and Residue Substitutions," *Protein Science* 11 (2002): 1393–1408.
70. I. D. Kuntz, J. M. Blaney, S. J. Oatley, R. Langridge, and T. E. Ferrin, "A Geometric Approach to Macromolecule-Ligand Interactions," *Journal of Molecular Biology* 161 (1982): 269–288.
71. D. Schneidman-Duhovny, Y. Inbar, R. Nussinov, and H. J. Wolfson, "PatchDock and SymmDock: Servers for Rigid and Symmetric Docking," *Nucleic Acids Research* 33 (2005): W363–W367.
72. M. Estrin and H. J. Wolfson, "SnapDock—Template-Based Docking by Geometric Hashing," *Bioinformatics* 33 (2017): i30–i36.
73. J. Esquivel-Rodríguez, Y. D. Yang, and D. Kihara, "Multi-LZerD: Multiple Protein Docking for Asymmetric Complexes," *Proteins* 80 (2012): 1818–1833.
74. C. Christoffer, S. Chen, V. Bharadwaj, et al., "LZerD Webserver for Pairwise and Multiple Protein-Protein Docking," *Nucleic Acids Research* 49 (2021): W359–W365.
75. E. J. Gardiner, P. Willett, and P. J. Artymiuk, "Protein Docking Using a Genetic Algorithm," *Proteins* 44 (2001): 44–56.
76. M. L. Connolly, "Shape Complementarity at the Hemoglobin $\alpha 1\beta 1$ Subunit Interface," *Biopolymers* 25 (1986): 1229–1247.
77. J. J. Gray, S. Moughon, C. Wang, et al., "Protein-Protein Docking With Simultaneous Optimization of Rigid-Body Displacement and Side-Chain Conformations," *Journal of Molecular Biology* 331 (2003): 281–299.
78. S. Chaudhury and J. J. Gray, "Conformer Selection and Induced Fit in Flexible Backbone Protein-Protein Docking Using Computational and NMR Ensembles," *Journal of Molecular Biology* 381 (2008): 1068–1087.
79. K. Bastard, C. Prévost, and M. Zacharias, "Accounting for Loop Flexibility During Protein-Protein Docking," *Proteins* 62 (2006): 956–969.
80. C. Wang, P. Bradley, and D. Baker, "Protein-Protein Docking With Backbone Flexibility," *Journal of Molecular Biology* 373 (2007): 503–519.
81. J. Fernández-Recio, M. Totrov, and R. Abagyan, "ICM-DISCO Docking by Global Energy Optimization With Fully Flexible Side-Chains," *Proteins* 52 (2003): 113–117.
82. S. Chaudhury, M. Berrondo, B. D. Weitzner, P. Muthu, H. Bergman, and J. J. Gray, "Benchmarking and Analysis of Protein Docking Performance in Rosetta v3. 2," *PLoS One* 6 (2011): e22477.
83. M. Zacharias, "ATTRACT: Protein-Protein Docking in CAPRI Using a Reduced Protein Model," *Proteins* 60 (2005): 252–256.
84. M. Kurcinski, S. Kmiecik, M. Zalewski, and A. Kolinski, "Protein-Protein Docking With Large-Scale Backbone Flexibility Using Coarse-Grained Monte-Carlo Simulations," *International Journal of Molecular Sciences* 22 (2021): 7341.
85. J. R. Sampson, *Adaptation in Natural and Artificial Systems* (Society for Industrial and Applied Mathematics, 1976).
86. C. Dominguez, R. Boelens, and A. M. Bonvin, "HADDOCK: A Protein-Protein Docking Approach Based on Biochemical or Biophysical Information," *Journal of the American Chemical Society* 125 (2003): 1731–1737.
87. M. Kurcinski, A. Badaczewska-Dawid, M. Kolinski, A. Kolinski, and S. Kmiecik, "Flexible Docking of Peptides to Proteins Using CABS-Dock," *Protein Science* 29 (2020): 211–222.
88. I. H. Moal and P. A. Bates, "SwarmDock and the Use of Normal Modes in Protein-Protein Docking," *International Journal of Molecular Sciences* 11 (2010): 3623–3648.
89. B. Jiménez-García, J. Roel-Touris, M. Romero-Durana, M. Vidal, D. Jiménez-González, and J. Fernández-Recio, "LightDock: A New Multi-Scale Approach to Protein-Protein Docking," *Bioinformatics* 34 (2018): 49–55.
90. S. Sunny and P. B. Jayaraj, "FPDock: Protein-Protein Docking Using Flower Pollination Algorithm," *Computational Biology and Chemistry* 93 (2021): 107518.
91. S. J. de Vries, M. van Dijk, and A. M. Bonvin, "The HADDOCK Web Server for Data-Driven Biomolecular Docking," *Nature Protocols* 5 (2010): 883–897.
92. S. J. de Vries, A. D. J. van Dijk, M. Krzeminski, et al., "HADDOCK Versus HADDOCK: New Features and Performance of HADDOCK2.0 on the CAPRI Targets," *Proteins* 69 (2007): 726–733.

93. A. Szilagyi and Y. Zhang, "Template-Based Structure Modeling of Protein-Protein Interactions," *Current Opinion in Structural Biology* 24 (2014): 10–23.
94. P. J. Kundrotas, Z. Zhu, J. Janin, and I. A. Vakser, "Templates Are Available to Model Nearly all Complexes of Structurally Characterized Proteins," *Proceedings of the National Academy of Sciences of the United States of America* 109 (2012): 9438–9441.
95. Q. C. Zhang, D. Petrey, L. Deng, et al., "Structure-Based Prediction of Protein-Protein Interactions on a Genome-Wide Scale," *Nature* 490, no. 7421 (2012): 556–560, <https://doi.org/10.1038/nature11503>.
96. P. J. Kundrotas and E. Alexov, "Predicting 3D Structures of Transient Protein-Protein Complexes by Homology," *Biochimica et Biophysica Acta (BBA) - Proteins and Proteomics* 1764 (2006): 1498–1511.
97. P. J. Kundrotas and E. Alexov, "PROTCOM: Searchable Database of Protein Complexes Enhanced With Domain-Domain Structures," *Nucleic Acids Research* 35 (2007): D575–D579.
98. P. J. Kundrotas, M. F. Lensink, and E. Alexov, "Homology-Based Modeling of 3D Structures of Protein-Protein Complexes Using Alignments of Modified Sequence Profiles," *International Journal of Biological Macromolecules* 43 (2008): 198–208.
99. S. Mukherjee and Y. Zhang, "Protein-Protein Complex Structure Predictions by Multimeric Threading and Template Recombination," *Structure* 19 (2011): 955–966.
100. A. Guerler, B. Govindarajoo, and Y. Zhang, "Mapping Monomeric Threading to Protein-Protein Structure Prediction," *Journal of Chemical Information and Modeling* 53 (2013): 717–725.
101. L. Lu, H. Lu, and J. Skolnick, "MULTIPROSPECTOR: An Algorithm for the Prediction of Protein-Protein Interactions by Multimeric Threading," *Proteins* 49 (2002): 350–364.
102. A. Baspinar, E. Cukuroglu, R. Nussinov, O. Keskin, and A. Gursoy, "PRISM: A Web Server and Repository for Prediction of Protein-Protein Interactions and Modeling Their 3D Complexes," *Nucleic Acids Research* 42 (2014): W285–W289.
103. R. Sinha, P. J. Kundrotas, and I. A. Vakser, "Docking by Structural Similarity at Protein-Protein Interfaces," *Proteins* 78 (2010): 3235–3241.
104. N. Tuncbag, O. Keskin, R. Nussinov, and A. Gursoy, "Fast and Accurate Modeling of Protein-Protein Interactions by Combining Template-Interface-Based Docking With Flexible Refinement," *Proteins* 80 (2012): 1239–1249.
105. S. Huang and X. Zou, "An Iterative Knowledge-Based Scoring Function for Protein-Protein Recognition," *Proteins* 72 (2008): 557–579.
106. L. C. Xue, J. P. Rodrigues, D. Dobbs, V. Honavar, and A. M. J. J. Bonvin, "Template-Based Protein-Protein Docking Exploiting Pairwise Interfacial Residue Restraints," *Briefings in Bioinformatics* 18 (2017): 458–466.
107. H. M. Berman, J. Westbrook, Z. Feng, et al., "The Protein Data Bank," *Nucleic Acids Research* 28 (2000): 235–242.
108. S. F. Altschul, W. Gish, W. Miller, E. W. Myers, and D. J. Lipman, "Basic Local Alignment Search Tool," *Journal of Molecular Biology* 215 (1990): 403–410.
109. S. F. Altschul, T. L. Madden, A. A. Schäffer, et al., "Gapped BLAST and PSI-BLAST: A New Generation of Protein Database Search Programs," *Nucleic Acids Research* 25 (1997): 3389–3402.
110. Y. Zhang and J. Skolnick, "TM-Align: A Protein Structure Alignment Algorithm Based on the TM-Score," *Nucleic Acids Research* 33 (2005): 2302–2309.
111. A. W. Ghoorah, M.-D. Devignes, M. Smaïl-Tabbone, and D. W. Ritchie, "Spatial Clustering of Protein Binding Sites for Template Based Protein Docking," *Bioinformatics* 27 (2011): 2820–2827.
112. N. Tuncbag, A. Gursoy, R. Nussinov, and O. Keskin, "Predicting Protein-Protein Interactions on a Proteome Scale by Matching Evolutionary and Structural Similarities at Interfaces Using PRISM," *Nature Protocols* 6 (2011): 1341–1354.
113. C. Quignot, G. Postic, H. Bret, et al., "InterEvDock3: A Combined Template-Based and Free Docking Server With Increased Performance Through Explicit Modeling of Complex Homologs and Integration of Covariation-Based Contact Maps," *Nucleic Acids Research* 49 (2021): W277–W284.
114. J. Dapkūnas, V. Kairys, K. Olechnovič, and Č. Venclovas, "Template-Based Modeling of Diverse Protein Interactions in CAPRI Rounds 38–45," *Proteins* 88 (2020): 939–947.
115. S. Vangaveti, T. Vreven, Y. Zhang, and Z. Weng, "Integrating Ab Initio and Template-Based Algorithms for Protein-Protein Complex Structure Prediction," *Bioinformatics* 36, no. 3 (2020): 751–757, <https://doi.org/10.1093/bioinformatics/btz623>.
116. D. Sarkar and S. Saha, "Machine-Learning Techniques for the Prediction of Protein-Protein Interactions," *Journal of Biosciences* 44 (2019): 104.
117. F. Soleymani, E. Paquet, H. Viktor, W. Michalowski, and D. Spinello, "Protein-Protein Interaction Prediction With Deep Learning: A Comprehensive Review," *Computational and Structural Biotechnology Journal* 20 (2022): 5316–5341.
118. M. Lee, "Recent Advances in Deep Learning for Protein-Protein Interaction Analysis: A Comprehensive Review," *Molecules* 28 (2023): 5169.
119. J. A. Ward, C. Perez-Lopez, and C. Mayor-Ruiz, "Biophysical and Computational Approaches to Study Ternary Complexes: A 'Cooperative Relationship' to Rationalize Targeted Protein Degradation," *ChemBioChem* 24 (2023): e202300163.
120. M. H. Doran, E. Pavada, M. J. Rynkiewicz, et al., "Cryo-EM and Molecular Docking Shows Myosin Loop 4 Contacts Actin and Tropomyosin on Thin Filaments," *Biophysical Journal* 119 (2020): 821–830.
121. D. Idrees and V. Kumar, "SARS-CoV-2 Spike Protein Interactions With Amyloidogenic Proteins: Potential Clues to Neurodegeneration," *Biochemical and Biophysical Research Communications* 554 (2021): 94–98.
122. I. T. Desta, K. A. Porter, B. Xia, D. Kozakov, and S. Vajda, "Performance and Its Limits in Rigid Body Protein-Protein Docking," *Structure* 28 (2020): 1071–1081.e3.
123. P. Gainza, F. Sverrisson, F. Monti, et al., "Deciphering Interaction Fingerprints From Protein Molecular Surfaces Using Geometric Deep Learning," *Nature Methods* 17 (2020): 184–192.
124. J. Tubiana, D. Schneidman-Duhovny, and H. J. Wolfson, "ScanNet: An Interpretable Geometric Deep Learning Model for Structure-Based Protein Binding Site Prediction," *Nature Methods* 19 (2022): 730–739.
125. E. Soltanikazemi, R. S. Roy, F. Quadir, et al., "Drlcomplex: Reconstruction of Protein Quaternary Structures Using Deep Reinforcement Learning," *arXiv* (2022) Preprint arXiv:220513594, <https://doi.org/10.48550/arXiv.2205.13594>.
126. F. Quadir, R. S. Roy, R. Halfmann, and J. Cheng, "DNCON2_Inter: Predicting Interchain Contacts for Homodimeric and Homomultimeric Protein Complexes Using Multiple Sequence Alignments of Monomers and Deep Learning," *Scientific Reports* 11 (2021): 12295.
127. Y. Yan and S.-Y. Huang, "Accurate Prediction of Inter-Protein Residue-Residue Contacts for Homo-Oligomeric Protein Complexes," *Briefings in Bioinformatics* 22 (2021): bbab038.
128. F. Sverrisson, J. Feydy, B. E. Correia, et al., "Fast End-To-End Learning on Protein Surfaces," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, (2021), 15272–15281.

129. Y. Wang, Y. Shen, S. Chen, et al., "Learning Harmonic Molecular Representations on Riemannian Manifold," *arXiv* (2023) Preprint arXiv:230315520, <https://doi.org/10.48550/arXiv.2303.15520>.
130. F. Monti, D. Boscaini, J. Masci, et al., "Geometric Deep Learning on Graphs and Manifolds Using Mixture Model CNNs," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, (2017), 5115–5124.
131. N. Keriven and G. Peyré, "Universal Invariant and Equivariant Graph Neural Networks," *Advances in Neural Information Processing Systems* 32 (2019): 7092–7101.
132. V. G. Satorras, E. Hoogeboom, and M. Welling, "E (n) Equivariant Graph Neural Networks," in *International Conference on Machine Learning (PMLR)*, (2021), 9323–9332.
133. S. Batzner, A. Musaelian, L. Sun, et al., "E (3)-Equivariant Graph Neural Networks for Data-Efficient and Accurate Interatomic Potentials," *Nature Communications* 13 (2022): 2453.
134. I. Batatia, S. Batzner, D. P. Kovács, et al., "The Design Space of E(3)-Equivariant Atom-Centered Interatomic Potentials," *arXiv* (2022) Preprint arXiv:220506643, <https://doi.org/10.48550/arXiv.2205.06643>.
135. N. Thomas, T. Smidt, S. Kearnes, et al., "Tensor Field Networks: Rotation and Translation-Equivariant Neural Networks for 3D Point Clouds," *arXiv* (2018) Preprint arXiv:180208219, <https://doi.org/10.48550/arXiv.1802.08219>.
136. F. Fuchs, D. Worrall, V. Fischer, et al., "Se (3)-Transformers: 3D Roto-Translation Equivariant Attention Networks," *Advances in Neural Information Processing Systems* 33 (2020): 1970–1981.
137. H. Stärk, O. Ganea, L. Pattanaik, et al., "Equibind: Geometric Deep Learning for Drug Binding Structure Prediction," in *International Conference on Machine Learning (PMLR)*, (2022), 20503–20521.
138. D. Cao, G. Chen, J. Jiang, et al., "Generic Protein–Ligand Interaction Scoring by Integrating Physical Prior Knowledge and Data Augmentation Modelling," *Nature Machine Intelligence* 6, no. 6 (2024): 688–700.
139. W. Kabsch, "A Solution for the Best Rotation to Relate Two Sets of Vectors," *Acta Crystallographica, Section A: Crystal Physics, Diffraction, Theoretical and General Crystallography* 32 (1976): 922–923.
140. W. Jin, R. Barzilay, and T. Jaakkola, "Antibody–Antigen Docking and Design via Hierarchical Structure Refinement," in *International Conference on Machine Learning (PMLR)*, (2022), 10217–10227.
141. Z. Yu, W. Huang, and Y. Liu, "Rigid Protein–Protein Docking via Equivariant Elliptic-Paraboloid Interface Prediction," *arXiv* (2024) Preprint arXiv:240108986, <https://doi.org/10.48550/arXiv.2401.08986>.
142. H. Wu, W. Liu, Y. Bian, et al., "EBMDock: Neural Probabilistic Protein–Protein Docking via a Differentiable Energy Model," in *The Twelfth International Conference on Learning Representations*.
143. Y. Ji, Y. Bian, G. Fu, et al., "Syndock: N Rigid Protein Docking via Learnable Group Synchronization," *arXiv* (2023) Preprint arXiv:230515156, <https://doi.org/10.48550/arXiv.2305.15156>.
144. G. Corso, H. Stärk, B. Jing, et al., "Diffdock: Diffusion Steps, Twists, and Turns for Molecular Docking," *arXiv* (2022). Preprint arXiv:221001776, <https://doi.org/10.48550/arXiv.2210.01776>.
145. J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan, et al., "Deep Unsupervised Learning Using Nonequilibrium Thermodynamics," in *International Conference on Machine Learning (PMLR)*, (2015), 2256–2265.
146. J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," *Advances in Neural Information Processing Systems* 33 (2020): 6840–6851.
147. Y. Song, J. Sohl-Dickstein, D. P. Kingma, et al., "Score-Based Generative Modeling Through Stochastic Differential Equations," *arXiv* (2020) Preprint arXiv:201113456, <https://doi.org/10.48550/arXiv.2011.13456>.
148. F. Sverrisson, M. Akdel, and D. Abramson, "DiffMaSIF: Surface-Based Protein–Protein Docking With Diffusion Models," in *Machine Learning in Structural Biology workshop at NeurIPS 2023*, (2023).
149. M. McPartlon, C. Marquet, T. Geffner, et al., "Latentdock: Protein–Protein Docking With Latent Diffusion," *MLSB* (2023).
150. V. R. Somnath, P. G. Sessa, M. R. Martinez, et al., "DockGame: Cooperative Games for Multimeric Rigid Protein Docking," *arXiv* (2023) Preprint arXiv:231006177, <https://doi.org/10.48550/arXiv.2310.06177>.
151. L.-S. Chu, S. Sarma, and J. J. Gray, "Unified Sampling and Ranking for Protein Docking With DFMDock," *bioRxiv* 2024 (2024): 615401, <https://doi.org/10.1101/2024.09.27.615401>.
152. N. Andrusier, E. Mashich, R. Nussinov, and H. J. Wolfson, "Principles of Flexible Protein–Protein Docking," *Proteins: Structure, Function, and Bioinformatics* 73 (2008): 271–289.
153. T. A. Hopf, C. P. Schärfe, J. P. Rodrigues, et al., "Sequence Co-Evolution Gives 3D Contacts and Structures of Protein Complexes," *eLife* 3 (2014): e03430, <https://doi.org/10.7554/eLife.03430>.
154. S. Ovchinnikov, H. Kamisetty, and D. Baker, "Robust and Accurate Prediction of Residue–Residue Interactions Across Protein Interfaces Using Evolutionary Information," *eLife* 3 (2014): e02030.
155. J. Ko and J. Lee, "Can AlphaFold2 Predict Protein–Peptide Complex Structures Accurately?," *bioRxiv* 2021 (2021): 453972.
156. T. Tsaban, J. K. Varga, O. Avraham, Z. Ben-Aharon, A. Khramushin, and O. Schueler-Furman, "Harnessing Protein Folding Neural Networks for Peptide–Protein Docking," *Nature Communications* 13 (2022): 176.
157. P. Bryant, G. Pozzati, and A. Elofsson, "Improved Prediction of Protein–Protein Interactions Using AlphaFold2," *Nature Communications* 13 (2022): 1265.
158. M. Gao, D. Nakajima An, J. M. Parks, and J. Skolnick, "AF2Complex Predicts Direct Physical Interactions in Multimeric Proteins With Deep Learning," *Nature Communications* 13 (2022): 1744.
159. R. M. Rao, J. Liu, R. Verkuil, et al., "MSA Transformer," in *International Conference on Machine Learning (PMLR)*, (2021), 8844–8856.
160. U. Lupo, D. Sgarbossa, and A.-F. Bitbol, "Protein Language Models Trained on Multiple Sequence Alignments Learn Phylogenetic Relationships," *Nature Communications* 13 (2022): 6298.
161. B. Chen, Z. Xie, J. Qiu, Z. Ye, J. Xu, and J. Tang, "Improved the Heterodimer Protein Complex Prediction With Protein Language Models," *Briefings in Bioinformatics* 24, no. 4 (2023): bbad221, <https://doi.org/10.1093/bib/bbad221>.
162. M. Mirdita, K. Schütze, Y. Moriwaki, L. Heo, S. Ovchinnikov, and M. Steinegger, "ColabFold: Making Protein Folding Accessible to all," *Nature Methods* 19 (2022): 679–682.
163. M. Baek, F. DiMaio, I. Anishchenko, et al., "Accurate Prediction of Protein Structures and Interactions Using a Three-Track Neural Network," *Science* 373, no. 6557 (2021): 871–876, <https://doi.org/10.1126/science.abj8754>.
164. C. Zhang, W. Zheng, S. M. Mortuza, Y. Li, and Y. Zhang, "DeepMSA: Constructing Deep Multiple Sequence Alignment to Improve Contact Prediction and Fold-Recognition for Distant-Homology Proteins," *Bioinformatics* 36 (2020): 2105–2112.
165. W. Zheng, Q. Wuyun, Y. Li, C. Zhang, L. Freddolino, and Y. Zhang, "Improving Deep Learning Protein Monomer and Complex Structure Prediction Using DeepMSA2 With Huge Metagenomics Data," *Nature Methods* 21 (2024): 279–289.
166. J. Zhu, Z. He, Z. Li, et al., "Uni-Fold MuSSe: De Novo Protein Complex Prediction With Protein Language Models," *bioRxiv* 2023 (2023): 528571.

167. W. Wang, Z. Peng, and J. Yang, "Single-Sequence Protein Structure Prediction Using Supervised Transformer Protein Language Models," *Nature Computational Science* 2 (2022): 804–814.
168. Z. Lin, H. Akin, R. Rao, et al., "Evolutionary-Scale Prediction of Atomic-Level Protein Structure With a Language Model," *Science* 379 (2023): 1123–1130.
169. U. Ghani, I. Desta, A. Jindal, et al., "Improved Docking of Protein Models by a Combination of AlphaFold2 and Cluspro," *bioRxiv* 2021 (2021): 459290.
170. A. Harmalkar, S. Lyskov, and J. J. Gray, "Reliable Protein-Protein Docking With AlphaFold, Rosetta, and Replica-Exchange," *bioRxiv* 2023 (2025): 551063, <https://doi.org/10.1101/2023.07.28.551063>.
171. P. Bryant, G. Pozzati, W. Zhu, A. Shenoy, P. Kundrotas, and A. Elofsson, "Predicting the Structure of Large Protein Complexes Using AlphaFold and Monte Carlo Tree Search," *Nature Communications* 13 (2022): 6028.
172. B. Shor and D. Schneidman-Duhovny, "CombFold: Predicting Structures of Large Protein Assemblies Using a Combinatorial Assembly Algorithm and AlphaFold," *Nature Methods* 21 (2024): 477–487.
173. Y. Luo, S. Li, Y. Sun, et al., "xTrimoDock: Rigid Protein Docking via Cross-Modal Representation Learning and Spectral Algorithm," *bioRxiv* 2023 (2023): 527251.
174. R. Wang, Y. Sun, Y. Luo, et al., "Injecting Multimodal Information Into Rigid Protein Docking via Bi-Level Optimization," *Advances in Neural Information Processing Systems* 36 (2023): 38266–38283.
175. R. Chen, J. Mintseris, J. Janin, and Z. Weng, "A Protein-Protein Docking Benchmark," *Proteins* 52 (2003): 88–91.
176. J. Mintseris, K. Wiehe, B. Pierce, et al., "Protein-Protein Docking Benchmark 2.0: An Update," *Proteins* 60 (2005): 214–216.
177. H. Hwang, B. Pierce, J. Mintseris, J. Janin, and Z. Weng, "Protein-Protein Docking Benchmark Version 3.0," *Proteins* 73 (2008): 705–709.
178. H. Hwang, T. Vreven, J. Janin, and Z. Weng, "Protein-Protein Docking Benchmark Version 4.0," *Proteins* 78 (2010): 3111–3114.
179. T. Vreven, I. H. Moal, A. Vangone, et al., "Updates to the Integrated Protein-Protein Interaction Benchmarks: Docking Benchmark Version 5 and Affinity Benchmark Version 2," *Journal of Molecular Biology* 427 (2015): 3031–3041.
180. Y. Yan and S.-Y. Huang, "A Non-Redundant Benchmark for Symmetric Protein Docking," *Big Data Mining and Analytics* 2 (2019): 92–99.
181. J. Janin, K. Henrick, J. Moult, et al., "CAPRI: A Critical Assessment of Predicted Interactions," *Proteins: Structure, Function, and Bioinformatics* 52 (2003): 2–9.
182. R. Méndez, R. Lepplae, M. F. Lensink, and S. J. Wodak, "Assessment of CAPRI Predictions in Rounds 3–5 Shows Progress in Docking Procedures," *Proteins: Structure, Function, and Bioinformatics* 60 (2005): 150–169.
183. R. Townshend, R. Bedi, P. Suriana, et al., "End-To-End Learning on 3d Protein Structure for Interface Prediction," *Advances in Neural Information Processing Systems* 32 (2019): 15642–15651.
184. A. Morehead, C. Chen, A. Sedova, and J. Cheng, "Dips-Plus: The Enhanced Database of Interacting Protein Structures for Interface Prediction," *Scientific Data* 10 (2023): 509.
185. R. D. Finn, J. Clements, and S. R. Eddy, "HMMER Web Server: Interactive Sequence Similarity Searching," *Nucleic Acids Research* 39 (2011): W29–W37.
186. R. Wang, X. Fang, Y. Lu, and S. Wang, "The PDBbind Database: Collection of Binding Affinities for Protein-Ligand Complexes With Known Three-Dimensional Structures," *Journal of Medicinal Chemistry* 47 (2004): 2977–2980.
187. D. A. Cannon, L. Shan, Q. Du, et al., "Experimentally Guided Computational Antibody Affinity Maturation With De Novo Docking, Modelling and Rational Design," *PLoS Computational Biology* 15 (2019): e1006980.
188. D. Kuroda and J. J. Gray, "Shape Complementarity and Hydrogen Bond Preferences in Protein-Protein Interfaces: Implications for Antibody Modeling and Protein-Protein Docking," *Bioinformatics* 32 (2016): 2451–2456.
189. B. D. Weitzner, J. R. Jeliakov, S. Lyskov, et al., "Modeling and Docking of Antibody Structures With Rosetta," *Nature Protocols* 12 (2017): 401–416.
190. C. Schneider, M. I. Raybould, and C. M. Deane, "SAbDab in the Age of Biotherapeutics: Updates Including SAbDab-Nano, the Nanobody Structure Tracker," *Nucleic Acids Research* 50 (2022): D1368–D1372.
191. J. Dunbar, K. Krawczyk, J. Leem, et al., "SAbDab: The Structural Antibody Database," *Nucleic Acids Research* 42 (2014): D1140–D1146.
192. J. Adolf-Bryfogle, O. Kalyuzhnyi, M. Kubitz, et al., "RosettaAntibodyDesign (RABD): A General Framework for Computational Antibody Design," *PLoS Computational Biology* 14 (2018): e1006112.
193. P. J. Kundrotas, I. Anishchenko, T. Dauzhenka, et al., "Dockground: A Comprehensive Data Resource for Modeling of Protein Complexes," *Protein Science* 27 (2018): 172–181.
194. K. W. Collins, M. M. Copeland, I. Koththoff, A. Singh, P. J. Kundrotas, and I. A. Vakser, "Dockground Resource for Protein Recognition Studies," *Protein Science* 31 (2022): e4481.
195. J. Yu and R. Guerois, "PPI4DOCK: Large Scale Assessment of the Use of Homology Models in Free Docking Over More Than 1000 Realistic Targets," *Bioinformatics* 32 (2016): 3760–3767.
196. D. Kovtun, M. Akdel, A. Goncareenco, et al., "PINDER: The Protein Interaction Dataset and Evaluation Resource," *bioRxiv* 2024 (2024): 603980.
197. E. Kozlova, A. Valentin, A. Khadhraoui, et al., "ProteinFlow: A Python Library to Pre-Process Protein Structure Data for Deep Learning Applications," *bioRxiv* 2023 (2023): 559346.
198. S. Lyskov and J. J. Gray, "The RosettaDock Server for Local Protein-Protein Docking," *Nucleic Acids Research* 36 (2008): W233–W238.
199. Y. Zhang and J. Skolnick, "Scoring Function for Automated Assessment of Protein Structure Template Quality," *Proteins: Structure, Function, and Bioinformatics* 57 (2004): 702–710.
200. A. Zemla, "LGA: A Method for Finding 3D Similarities in Protein Structures," *Nucleic Acids Research* 31 (2003): 3370–3374.
201. M. Gao and J. Skolnick, "New Benchmark Metrics for Protein-Protein Docking Methods," *Proteins: Structure, Function, and Bioinformatics* 79 (2011): 1623–1634.
202. V. Mariani, M. Biasini, A. Barbato, and T. Schwede, "IDDT: A Local Superposition-Free Score for Comparing Protein Structures and Models Using Distance Difference Tests," *Bioinformatics* 29 (2013): 2722–2728.
203. S. Basu and B. Wallner, "DockQ: A Quality Measure for Protein-Protein Docking Models," *PLoS One* 11 (2016): e0161879.
204. B. Jiménez-García, C. Pons, D. I. Svergun, P. Bernadó, and J. Fernández-Recio, "pyDockSAXS: Protein-Protein Complex Structure by SAXS and Computational Docking," *Nucleic Acids Research* 43 (2015): W356–W361.
205. B. G. Pierce, K. Wiehe, H. Hwang, B. H. Kim, T. Vreven, and Z. Weng, "ZDOCK Server: Interactive Docking Prediction of Protein-Protein Complexes and Symmetric Multimers," *Bioinformatics* 30 (2014): 1771–1773.

206. G. Macindoe, L. Mavridis, V. Venkatraman, M. D. Devignes, and D. W. Ritchie, "HexServer: An FFT-Based Protein Docking Server Powered by Graphics Processors," *Nucleic Acids Research* 38 (2010): W445–W449.
207. V. I. Lesk and M. J. E. Sternberg, "3D-Garden: A System for Modelling Protein–Protein Complexes Based on Conformational Refinement of Ensembles Generated With the Marching Cubes Algorithm," *Bioinformatics* 24 (2008): 1137–1144.
208. G. Weng, E. Wang, Z. Wang, et al., "HawkDock: A Web Server to Predict and Analyze the Protein–Protein Complex Based on Computational Docking and MM/GBSA," *Nucleic Acids Research* 47 (2019): W322–W330.
209. M. Torchala, I. H. Moal, R. A. G. Chaleil, J. Fernandez-Recio, and P. A. Bates, "SwarmDock: A Server for Flexible Protein–Protein Docking," *Bioinformatics* 29 (2013): 807–809.
210. D. Schneidman-Duhovny, M. Hammel, and A. Sali, "Macromolecular Docking Restrained by a Small Angle X-Ray Scattering Profile," *Journal of Structural Biology* 173 (2011): 461–471.
211. D. Schneidman-Duhovny, M. Hammel, J. A. Tainer, and A. Sali, "FoXS, FoXSDock and MultiFoXS: Single-State and Multi-State Structural Modeling of Proteins and Their Complexes Based on SAXS Profiles," *Nucleic Acids Research* 44 (2016): W424–W429.
212. C. Quignot, J. Rey, J. Yu, P. Tufféry, R. Guerois, and J. Andreani, "InterEvDock2: An Expanded Server for Protein Docking Using Evolutionary and Biological Information From Homology Models and Multimeric Inputs," *Nucleic Acids Research* 46 (2018): W408–W416.
213. A. Tovchigrechko and I. A. Vakser, "GRAMM-X Public Web Server for Protein–Protein Docking," *Nucleic Acids Research* 34 (2006): W310–W314.
214. A. Singh, M. M. Copeland, P. J. Kundrotas, et al., "GRAMM Web Server for Protein Docking," in *Computational Drug Discovery and Design* (Springer, 2023), 101–112.
215. D. S. Goodsell and A. J. Olson, "Structural Symmetry and Protein Function," *Annual Review of Biophysics and Biomolecular Structure* 29 (2000): 105–153.
216. I. André, C. E. Strauss, D. B. Kaplan, C. E. M. Strauss, P. Bradley, and D. Baker, "Emergence of Symmetry in Homooligomeric Biological Assemblies," *Proceedings of the National Academy of Sciences of the United States of America* 105, no. 42 (2008): 16148–16152, <https://doi.org/10.1073/pnas.0807576105>.
217. B. Han, C. Ren, W. Wang, J. Li, and X. Gong, "Computational Prediction of Protein Intrinsically Disordered Region Related Interactions and Functions," *Genes* 14 (2023): 432.
218. A. S. Holehouse and B. B. Kragelund, "The Molecular Basis for Cellular Function of Intrinsically Disordered Protein Regions," *Nature Reviews Molecular Cell Biology* 25 (2024): 187–211.
219. F. Gaudreault, C. R. Corbeil, and T. Sulea, "Enhanced Antibody–Antigen Structure Prediction From Molecular Docking Using AlphaFold2," *Scientific Reports* 13 (2023): 15107.
220. S.-Y. Huang and X. Zou, "A Knowledge-Based Scoring Function for Protein–RNA Interactions Derived From a Statistical Mechanics-Based Iterative Method," *Nucleic Acids Research* 42 (2014): e55.
221. A. Kryshtafovych, T. Schwede, M. Topf, K. Fidelis, and J. Moult, "Critical Assessment of Methods of Protein Structure Prediction (CASP)—Round XIV," *Proteins: Structure, Function, and Bioinformatics* 89 (2021): 1607–1617.
222. A. Gangwal, A. Ansari, I. Ahmad, A. K. Azad, and W. M. A. Wan Sulaiman, "Current Strategies to Address Data Scarcity in Artificial Intelligence-Based Drug Discovery: A Comprehensive Review," *Computers in Biology and Medicine* 179 (2024): 108734.
223. Z. Guo, C. Zhang, W. Yu, et al., "Few-Shot Graph Learning for Molecular Property Prediction," in *Proceedings of the Web Conference 2021*, (2021), 2559–2567.
224. H. Xu, Y. Chen, and D. Zhang, "Worth of Prior Knowledge for Enhancing Deep Learning," *Nexus* 1 (2024): 100003, <https://doi.org/10.1016/j.jynexs.2024.100003>.